Climate Change, Comparative Advantages, and Structural Transformation in Agriculture

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Abstract

Climate change is expected to reshape agricultural productivity and comparative advantages across countries, with important consequences for economic welfare. Yet these consequences depend not only on where productivity changes, but also on how countries consume, produce, and trade. This paper investigates how consumption patterns and international trade shape the unequal welfare consequences of climate change. I combine spatially disaggregated data on crop-specific agricultural productivity under alternative climate conditions with bilateral trade flows and country-level expenditure data in a multi-country Ricardian general equilibrium trade model with non-homothetic preferences. I find that climate-induced productivity shocks reduce global welfare by 0.83% of world GDP, with particularly large losses in lower-middle-income economies. A welfare decomposition reveals that the price effect dominates the structural change effect, which captures income-driven adjustments in consumption expenditure shares. Counterfactual simulations further show that differences in food expenditure shares amplify welfare losses in poorer economies. Finally, export restrictions by major agricultural suppliers exacerbate these losses, while removing export taxes can partially mitigate them. These findings highlight how consumption exposure and international trade policies shape the unequal economic consequences of climate change.

JEL Classification: F13, F18, F40, Q11, Q17, Q54

Keywords: International Trade, Welfare, Trade and Environment, Agricultural Commodities, Farming, Climate Change.

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1. Introduction

Climate change has the ability to increase average temperatures and the frequency of climate extremes (Leonard et al., 2014; AghaKouchak et al., 2014, 2020). The predictions of the Intergovernmental Panel on Climate Change (IPCC) warn against a worsening of the situation in the coming years. This results in an increase in the frequency, intensity, spatial extent, and duration of extreme and increasingly complex climatic events (Bathke et al., 2015; Tavakol et al., 2020) in all regions in the world. The impact of climate change on agriculture is particularly pronounced, showing substantial disparities in various geographical areas. Food security is affected by uncertainty over domestic production and production in foreign countries. The resilience of agri-food supply chains to climate change is also of great concern.

Crop producers are exposed to climatic risk as well as other forms of risk (e.g., market risk), which leads them to combine several business and financial risk management tools. Because crops serve multiple end uses, climate hazards affect the production of human food, animal feed, and biofuels. The competitiveness of animal production is largely dependent on feed costs and hence on the performance of the grain sector. Canada and the United States are major exporters of field crops and meat, while most African countries and some Asian countries are net importers. In each of these countries, food security depends on the ability to import. The fluidity of trade and the accessibility of cereals and meat are fundamental to national and international food security.

Due to its heterogeneous impacts, climate change affects the competitiveness of firms and countries' comparative advantage inducing changes in specializations between countries and adjustments in trade flows (Costinot et al., 2016; Nath, 2020, 2025). As comparative advantages evolve over time (Gaigné and Gouel, 2022), climate change and shifting patterns of specialization are likely to have significant implications for welfare and consumption baskets. Empirical studies using agro-ecological characteristics also document substantial geographical differences in the effects of climate change on agricultural productivity and economic outcomes (Fezzi and Bateman, 2015; Chalise and Naranpanawa, 2016; Gebreegziabher et al., 2020; Hossain et al., 2019). If climate change impacts all crops and countries uniformly, the resulting reduction in cross-country productivity differences may weaken trade incentives, leading to substantial welfare losses (Reilly and Hohmann, 1993; Randhir and Hertel, 2000). By contrast, if climate change increases productivity disparities across regions, it could amplify trade opportunities and, in doing so, help mitigate some of the welfare losses (Gouel and Laborde, 2021; Nath, 2025; Lamonaca et al., 2024). This paper addresses these challenges by examining the macro-level consequences of climate change. My main contribution to the existing literature is twofold.

First, I show how income-dependent food demand shapes the welfare consequences of climate change. The existing literature has largely relied on homothetic or quasi-linear preferences (Costinot et al., 2016; Gouel and Laborde, 2021), which suppress the feedback between income losses and food expenditure shares. The assumption of homotheticity is innocuous when productivity shocks are small and symmetric. However, it becomes difficult to sustain in the context of climate change, which disproportionately lowers agricultural productivity, and therefore real

income, in countries where households already devote a large share of their budget to food (Uy et al., 2013; Herrendorf et al., 2014). To address this limitation, I combine the Ricardian model of agricultural trade of Costinot et al. (2016) with the non-homothetic CES demand system of Comin et al. (2021). This choice of preferences is disciplined by three requirements that alternative specifications fail to meet jointly: (*i*) the system accommodates good-specific income elasticities, thereby allowing agricultural and non-agricultural goods to exhibit empirically distinct Engel curves; (*ii*) it preserves a constant elasticity of substitution across goods, which ensures tractability in general equilibrium; (*iii*) it delivers a closed-form expression for the consumer price index in terms of observable expenditure shares and prices, rather than unobservable utility levels, which is crucial for conducting consistent welfare comparisons within a quantitative trade model. I estimate the demand system using cross-country data and obtain an income elasticity of food demand of approximately 0.55, consistent with Engel's law. I calibrate countries' exposure to food prices using observed agricultural expenditure shares and incorporate the field-level productivity changes from Costinot et al. (2016). These climate-induced productivity shocks vary substantially across countries and crops, with an aggregate loss equivalent to 0.34% of world GDP. Using these shocks, I find a global welfare loss of 0.83% of GDP, compared with 0.74% under homothetic preferences and 0.26% in Costinot et al. (2016). On average, welfare losses exceed 6% of income in both low-income and lower-middle-income economies. Cross-country differences in food expenditure shares are central to this regressive pattern of climate shocks: assigning every country the same food share cuts the losses in low-income economies by roughly half. Food expenditure shares thus determine how exposed lower-income economies are, but not the full extent of their losses, which also depends on the size of the climate-induced rise in food prices and on the response of income.

Second, I examine the channels through which climate shocks affect welfare. I measure welfare using a money-metric utility approach (Baqaee and Burstein, 2023; Baqaee et al., 2024; Jaravel and Lashkari, 2024) that delivers consistent comparisons of the cost of living before and after climate change when preferences are non-homothetic. The money-metric utility (McKenzie, 1957; Samuelson, 1974) translates welfare changes into monetary units by evaluating the expenditure function at a reference price vector, delivering a cardinal welfare measure that is comparable across countries. This measure is directly related to the equivalent variation[1]. In the presence of non-homothetic preferences, I decompose the total welfare effect into an income effect, capturing changes in farm income; a price effect, capturing the welfare loss arising from climate-induced changes in relative prices of agricultural goods; and a structural change effect, reflecting shifts in the share of agricultural goods in total consumption. The results show that the price effect is the dominant channel of welfare transmission, exceeding the structural change effect across all income groups and specifications. This finding complements Nath (2025) by showing that, in addition to constraints on sectoral reallocation, the welfare incidence of climate change depends critically on food price changes and income-driven expenditure shares.

[1]To measure the aggregate welfare consequences of climate change, I adopt the equivalent variation in the original sense of Hicks, as clarified by Diewert (1992): $EV = e(p^0, u^1) - e(p^0, u^0)$, which evaluates the post-shock utility level u^1 at pre-shock prices p^0. Diewert (1992) notes that Hicks (1946) later used the alternative definition $H = e(p^1, u^1) - e(p^0, u^1)$. As demonstrated in Appendix H, the two measures have the same magnitude and opposite signs when income is unchanged, while EV additionally captures the change in income.

The analysis relies on spatially disaggregated agricultural productivity data from the Global Agro-Ecological Zones (Fischer et al., 2021) project of the Food and Agricultural Organization (FAO), which combines agronomic modeling with detailed geospatial information–including soil type, elevation, and climate related variables to estimate potential yields for individual crops worldwide. Comparing yields under present and future climate scenarios makes it possible to infer changes in comparative advantage across space and time. The results indicate that climate change has uneven effects on agri-food productivity and welfare, with low- and lower-middle-income economies especially vulnerable due to their reliance on climate-sensitive sectors. Exposure to climate shocks can have substantial effects on price variation compared to structural changes in demand shares for agricultural products.

In an application, I use the framework to examine how trade policies shape the welfare consequences of climate change. While Costinot et al. (2016) and Gouel and Laborde (2021) emphasize the role of international trade in facilitating adaptation to climate change, recent evidence from Hsiao et al. (2026) highlights how governments adjust agricultural trade policies in response to climate shocks. Motivated by this evidence, I conduct counterfactual experiments to assess how export restrictions by major agricultural suppliers affect welfare. I find that these restrictions exacerbate global welfare losses, whereas removing export taxes in a separate counterfactual can partially mitigate them. These results underscore how trade policies can influence the effectiveness of international trade as a mechanism of adaptation to climate change.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature on climate change, comparative advantage, and welfare measurement under non-homothetic preferences. Section 3 documents some key facts that motivate the analysis: the evolving nature of comparative advantages across countries and the heterogeneous exposure to climate shocks across income levels. Section 4 develops the theoretical framework, embedding non-homothetic CES preferences into a multi-country Ricardian trade model of agriculture with heterogeneous land. Section 5 describes the data, including the FAO Global Agro-Ecological Zones dataset, household expenditure data, bilateral trade flows, and tariff data. Section 6 brings the model to the data: it presents the estimation of the structural parameters, including the income elasticity of demand for agrifood products, the design of the climate counterfactuals, and the measurement of welfare. Section 7 reports the counterfactual results: it quantifies the welfare consequences of climate change, decomposes them into income, price, and structural change effects, assesses their sensitivity to alternative climate scenarios and demand specifications, examines adaptation, and simulates trade policy responses to climate shocks. Section 8 discusses the policy implications, and Section 9 concludes and outlines directions for future research.

2. A Brief Literature Review

This paper is related to three strands of the literature. The first examines how climate change affects agricultural productivity and comparative advantage through quantitative Ricardian trade models (Costinot et al., 2012, 2016; Gouel and Laborde, 2021). These models build on heterogeneous productivity across countries, crops, and locations, with the agricultural framework of

Costinot et al. (2016) allowing for land heterogeneity and yields drawn from a Fréchet distribution. Related studies examine the role of productivity and technological heterogeneity in shaping specialization and international trade (Huang et al., 2011; Heerman, 2020; Heerman and Sheldon, 2022; Levchenko and Zhang, 2013; Simonovska and Waugh, 2014; Caliendo and Parro, 2015). Because climate change reshapes comparative advantage differently across countries, international trade becomes an important margin of adjustment: countries experiencing productivity declines can source food from less affected regions, thereby mitigating welfare losses. This adaptation channel has been examined by Cui (2020), Conte et al. (2021), Gouel and Laborde (2021), and more recently Nath (2025). A recent review by Farrokhi et al. (2026) further discusses the role of production reallocation, specialization, and international trade in climate adaptation. Trade policies can, however, limit such adjustments. Reimer and Li (2009) show that trade restrictions amplify the consequences of yield variability, while export restrictions can further exacerbate food price increase (Martin and Anderson, 2012; Giordani et al., 2016). Gouel (2016) examines how trade policy coordination is affected by governments' incentives to stabilize domestic food prices. More recently, Hsiao et al. (2026) document how governments adjust agricultural trade policies in response to climate shocks, with regressive welfare consequences. The benchmark frameworks of Costinot et al. (2016) and Gouel and Laborde (2021), however, rely on quasi-linear preferences, which abstract from income effects on agricultural demand. Building on this literature, and using climate-induced productivity shocks at the field-level from Costinot et al. (2016), I examine how income-dependent food expenditure patterns shape the distribution of welfare losses across countries.

A second strand of literature examines structural transformation and non-homothetic demand. Economic development is associated with large reallocations of expenditure and production across agriculture, manufacturing, and services (Uy et al., 2013; Herrendorf et al., 2014). In the agri-food sector, changes in consumption patterns and comparative advantage jointly shape the evolution of agricultural trade and specialization (Gaigné and Gouel, 2022). In quantitative trade models, Fieler (2011) and Tombe (2015) highlight the importance of non-homothetic demand for bilateral trade patterns, agricultural specialization, and international productivity differences. More recently, Comin et al. (2021) introduce non-homothetic CES preferences into a multi-sector general equilibrium model and derive a tractable demand system in which expenditure shares respond to income and relative prices. Lewis et al. (2022) embed this demand structure into a quantitative trade model to study how shifts in consumption patterns affect global trade flows. While these studies focus on how income-dependent demand shapes long-run structural transformation and international trade, I examine how non-homothetic preferences influence the welfare consequences of climate-induced productivity shocks in agriculture. Particularly relevant is Nath (2025), who also allows food demand to be non-homothetic and shows that subsistence food needs and high trade barriers may intensify agricultural specialization in vulnerable low-income countries, limiting adaptation through sectoral reallocation. Whereas he emphasizes the supply-side margin of sectoral reallocation, this analysis takes a complementary approach by focusing on the consumption-side welfare channel, and by decomposing welfare changes using a money-metric measure that separates income, price, and structural change effects.

A third strand concerns welfare measurement under non-homothetic preferences. When consumption patterns vary with income, these differences must be taken into account to accurately measure changes in the cost of living (Baqaee and Burstein, 2023; Baqaee et al., 2024; Jaravel and Lashkari, 2024). Following the money-metric utility approach (McKenzie, 1957; Samuelson, 1974) and the equivalent variation framework (Hicks, 1946; Diewert, 1992), I measure the welfare consequences of climate change using the expenditure function and decompose them into income, price, and structural change effects. This decomposition distinguishes the welfare implications of relative price changes from those associated with endogenous adjustments in expenditure shares, while accounting for differences in initial food expenditure exposure across countries.

3. Stylized Facts

The analysis is based on two key insights: (*i*) the observable and evolving nature of comparative advantages across countries and (*ii*) the fact that the impact of climate change vary with consumer income levels

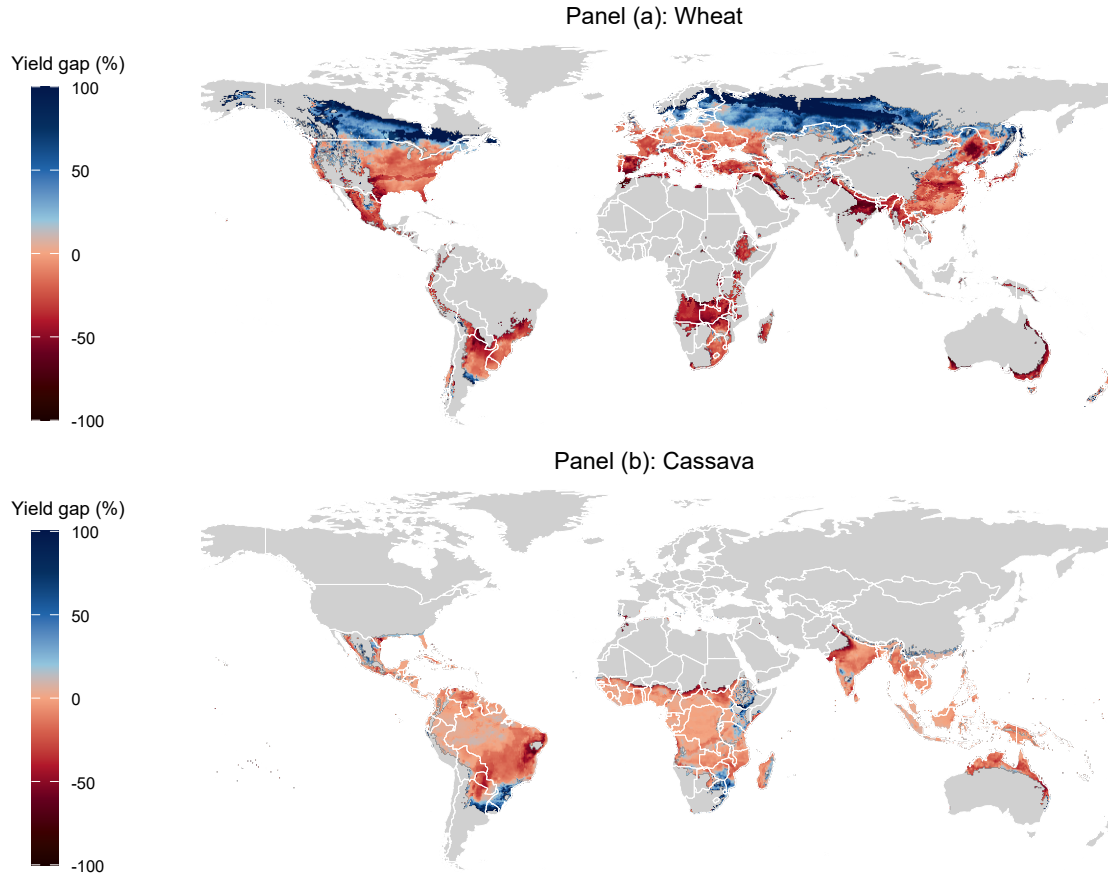
3.1. Evolving comparative advantages

In a globalized world, the impact of micro-level shocks is determined not only by their average magnitude but also by their spatial dispersion, especially in agriculture. If climate change affects agricultural productivity, it will, in turn, influence specialization patterns and countries' ability to benefit from trade, ultimately impacting consumer welfare. I define comparative advantages by a country's ability to specialize in producing goods for which it has relatively higher productivity or lower opportunity cost, allowing more efficient global trade (Donaldson, 2019). In line with standard Ricardian frameworks, cross-country differences in productivity are the primary drivers of comparative advantage, and these differences evolve endogenously with climate change.

Figure 1 documents this evolution by plotting, for each 5-arcminute grid cell, the percentage change in relative productivity induced by climate shocks. For each grid-cell, I compute the ratio of post-shock to baseline yield, scaled by the corresponding world averages, and express the result as a percentage deviation from zero. A positive (negative) value indicates that the cell gains (loses) comparative advantage relative to the world average.

Panel (a) indicates a reallocation of comparative advantage in wheat production. High-latitude economies, Canada, Russia, and Northern Europe, gain substantially in relative productivity as warmer temperatures expand viable growing areas northward. By contrast, wheat-producing regions in the Middle East, South Asia, and sub-Saharan Africa experience pronounced deterioration in their relative position, reflecting the sensitivity of wheat yields to extreme heat. Panel (b) documents a symmetric pattern for cassava, whose production is concentrated in low-income tropical economies. Across sub-Saharan Africa, South Asia, and Latin America, climate change reduces relative productivity with limited compensating gains elsewhere. These

Figure 1: Climate-induced changes in comparative advantage for wheat and cassava.



Notes: The figure plots, for each 5-arcminute grid cell, the percentage change in relative productivity induced by climate change, computed as the ratio of post-shock to baseline yield scaled by the corresponding world averages, minus one. Red (blue) areas indicate a deterioration (improvement) in comparative advantage relative to the world average. Grey areas correspond to cells with zero yield under the baseline scenario. Climate scenarios compare potential yields under RCP 8.5 (2041–2070) to a 1981–2010 historical baseline, assuming rain-fed cultivation at intermediate input levels. *Source:* Author's calculations using the Global Agro-Ecological Zones (GAEZ v4) database (Fischer et al., 2021), Food and Agriculture Organization (FAO).

are economies where food expenditure shares are largest, making the erosion of agricultural comparative advantage particularly consequential for household welfare.

3.2. Structural Change, exposure to shocks, and income heterogeneity

To move beyond the analysis of comparative advantage and quantify the macroeconomic consequences of climate change, this paper connects the distributional impact of climate shocks on household welfare to income levels.

In Figure 2, I illustrate sectoral value-added shares relative to GDP per capita in 43 countries from 2000 to 2014, indicating three key patterns of structural change[2]: (1) a decline in agriculture's share as economies develop, (2) a hump-shaped trajectory of industry's share, and (3) an increasing dominance of the services sector. The increase in GDP per capita is associated with a decrease in the nominal value added share in agriculture. Previous research provided extensive documentation of these trends across different sets of countries and time horizons (e.g., Buera and Kaboski (2012); Herrendorf et al. (2014)). In many low-income countries, agriculture employs a large share of the population but contributes a small share of GDP. As a result, climate shocks to agriculture are expected to reduce both labor income and food availability for a substantial portion of households. This paper explores these issues and shows that accounting for preferences exacerbates the impact of climate shocks on welfare. Poor households spend a larger share of income on food compared to wealthier households, and climate-induced increases in food prices are expected to lead to a significant decline in real purchasing power. Thus, even small GDP losses can translate into large utility losses for low-income populations.

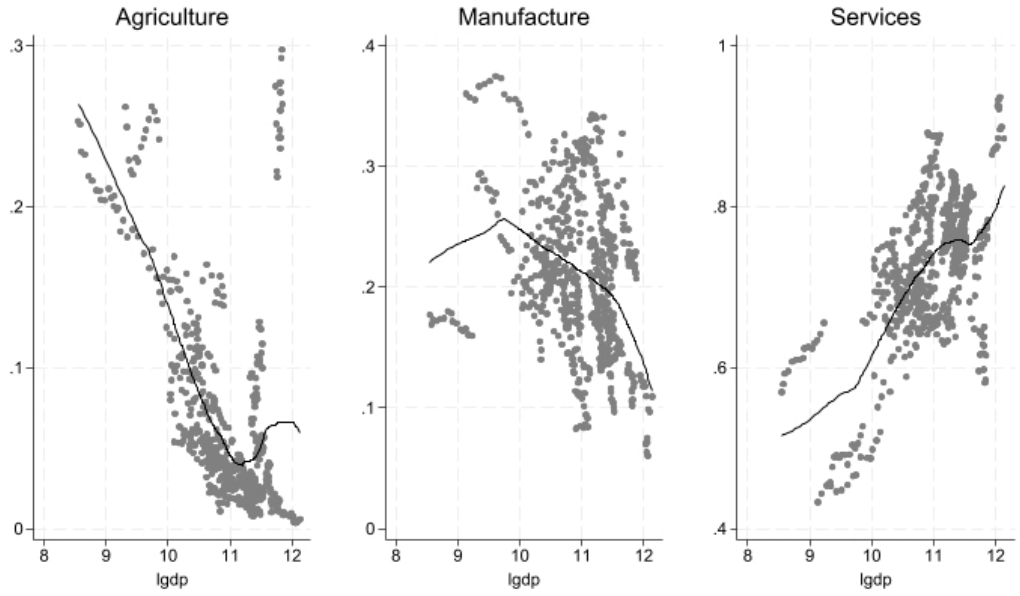
In addition to these facts, and using the World Input-Output Database (WIOD), I measure (a) the share of value added in gross output for each sector, and (b) the share of each sector's use of intermediate input in every other sector (including its own). Table 1 reports these shares for the five richest and five poorest countries in the sample-classified by GDP per capita—focusing on the most recent year available, 2024.

For agriculture, the results indicate a higher share of value-added in gross output for low-income countries, which implies that high-income countries use intermediate inputs more intensively than low-income countries. However, within the agricultural sector, low-income countries rely more on agricultural inputs (for e.g, seeds and animal feed), whereas high-income countries incorporate a larger share of service-based inputs. The key takeaways from this fact are: (i) the share of value added in agricultural output tends to be higher in vulnerable low-income countries than in high-income countries, and (ii) low-income countries are more exposed to micro-shocks affecting the use of inputs in agriculture.

In sum, climate shocks affects agricultural productivity unevenly across space, with the largest

[2] As in Buera and Kaboski (2012); Herrendorf et al. (2014), I define structural change as the reallocation of economic activity across three broad sectors (agriculture, manufacturing, and services) following the process of modern economic growth.

Figure 2: Sectoral shares in value added



Notes: The sample covers 44 countries over the period 2000–2014. Each subfigure plots a sector's share in aggregate value added (vertical axis) against log GDP per capita (horizontal axis). Value added is measured at current prices; GDP per capita is measured using expenditure-side real GDP at chained PPPs. *Source:* World Input-Output Database (WIOD).

losses concentrated in economies where agriculture represents the largest share of output and household budgets. Although documented country-level patterns confirm that lower-income economies are more agriculture-dependent, the substantial dispersion visible within income groups suggests that aggregate data alone cannot identify which countries or crops are the most exposed. The high-resolution spatial data compiled in this paper reveal that comparative advantage deteriorates most sharply in precisely those economies, where food expenditure shares are already the highest. The following sections embed these patterns into a general equilibrium Ricardian model with non-homothetic preferences to quantify how income effects shape the aggregate and distributional welfare consequences of climate change.

4. Theoretical framework

I describe a trade model in which the world consists of multiple importing countries j and exporting countries i that are endowed with land L_i and labor N_i. Following Eaton and Kortum (2002), I assume that countries produce and trade multiple crops indexed by $k \in \{1, \ldots, K\}$ and an outside good (which is a composite of manufactured goods and services). Labor can be freely allocated across crops and sectors, perfectly mobile within a country, and immobile across countries. Land consists of various fields, indexed by $f \in \mathcal{F}_i$, and each field includes a continuum of diverse parcels, indexed by $\omega \in [0, 1]$. All fields in the dataset correspond to 5-arcminute grid cells as in Costinot et al. (2016). Land is allocated across crop production activities. I further assume that countries' comparative advantage is determined not only by their productivity (Heerman, 2020), but also by their agro-ecological characteristics and their

Table 1: Input intensities for Rich and Poor Countries

	low-income countries (high-income countries)		
	Output sector		
Input Sector	**Agriculture**	**Manufacturing**	**Services**
Agriculture	0.35 (0.30)	0.12 (0.07)	0.02 (0.07)
Manufacturing	0.41 (0.33)	0.49 (0.48)	0.14 (0.27)
Services	0.24 (0.37)	0.39 (0.45)	0.84 (0.66)
Value Added Shares	0.68 (0.46)	0.32 (0.33)	0.59 (0.57)

Notes: Country groups are defined based on GDP per capita: "high-income" ("low-income") refers to the average of the 5 wealthiest (poorest) economies in the sample in term of GDP. For each output sector, intermediate input shares reflect the proportion of total intermediate expenditures allocated to each input sector. Value-added shares represent value added as a fraction of gross output for each sector. *Source:* Author's calculations based on the World Input-Output Database (WIOD).

trade costs. Following Matsuyama (2019) and Comin et al. (2021), I assume that consumers have non-homothetic preferences, such that the demand for agricultural goods varies with the level of income at given prices.

Climate change is modeled through changes in crop-specific productivity. This affects crop supply, producer prices, trade flows, and consumer prices. Under non-homothetic preferences, the welfare effect of these price changes depends on income: poorer countries devote a larger share of expenditure to food and are therefore more exposed to agricultural price shocks.

4.1. Demand

I model demand through a three-stage budgeting framework. In the first stage, a representative household allocates expenditure between an agricultural aggregate C_j^a and a non-agricultural (outside) good C_j^o. The non-agricultural good C_j^o is a homogeneous composite of manufacturing and services. It is freely traded at zero cost and serves as the numéraire ($P_j^o = 1$), and all non-homotheticity in the model operates through the agricultural aggregate (Comin et al., 2021; Tombe, 2015; Matsuyama, 2019). In the second stage, expenditure on the agricultural aggregate is allocated across crops. In the third stage, expenditure on each crop is allocated across varieties from different origins.

Stage 1: Aggregate demand. In each country, I define a non-homothetic CES utility function for a representative agent, deriving utility from the consumption of an outside good C_j^o and an aggregator of all crops, C_j^a. Let N_j denote population, which equals the labor endowment, and define aggregate real consumption per capita as:

$$U_j \equiv \frac{C_j}{N_j} = \frac{E_j}{P_j N_j}, \tag{1}$$

where E_j is the total expenditure and P_j is the ideal consumer price index. Following Hanoch (1975) and Comin et al. (2021), preferences are represented by a non-homothetic CES aggregator:

$$\beta_a^{1/\alpha}\, U_j^{\varepsilon_a \frac{1-\alpha}{\alpha}} \left(\frac{C_j^a}{N_j}\right)^{\frac{\alpha-1}{\alpha}} + \beta_o^{1/\alpha}\, U_j^{\varepsilon_o \frac{1-\alpha}{\alpha}} \left(\frac{C_j^o}{N_j}\right)^{\frac{\alpha-1}{\alpha}} = 1. \tag{2}$$

Here, β_a and β_o govern the relative importance of agricultural and non-agricultural goods, with $\beta_a + \beta_o = 1$. The parameter $\alpha > 0$ is the elasticity of substitution between the two goods, while ε_a and ε_o govern how the demand for each good varies with real income. The homothetic CES case is obtained when $\varepsilon_a = \varepsilon_o = 1$. Climate-induced productivity shocks affect welfare not only through prices, trade, and comparative advantage, but also through income-driven changes in food expenditure shares. The representative household maximizes utility subject to the budget constraint:

$$E_j = P_j^a C_j^a + P_j^o C_j^o = w_j N_j + r_j L_j + T_j + D_j, \tag{3}$$

where P_j^a and P_j^o are the prices of the agricultural composite and the non-agricultural good, respectively. Total income is the sum of labor income, $w_j N_j$, where w_j is the wage and N_j the labor endowment; land income, $r_j L_j$, where L_j is the land endowment and r_j the average land rent across fields; trade-tax revenue, T_j, rebated lump-sum to the household; and the trade deficit D_j, with $\sum_j D_j = 0$. Solving the household's problem (proof in Appendix A) yields the optimal expenditure share on good $b \in \{a, o\}$:

$$(\beta_b)^{\frac{1}{\alpha}} \left(\frac{C_j}{N_j}\right)^{\frac{\varepsilon_b(1-\alpha)}{\alpha}} \left(\frac{C_j^b}{N_j}\right)^{\frac{\alpha-1}{\alpha}} = \frac{P_j^b C_j^b}{E_j}. \tag{4}$$

The optimal consumption C_j^b of good b can then be written as:

$$C_j^b = \beta_b N_j \left(\frac{P_j^b N_j}{E_j}\right)^{-\alpha} \left(\frac{C_j}{N_j}\right)^{\varepsilon_b(1-\alpha)}. \tag{5}$$

Since $E_j = P_j C_j$, equation (5) can equivalently be written as

$$C_j^b = \beta_b N_j \left(\frac{P_j^b}{P_j}\right)^{-\alpha} \left(\frac{C_j}{N_j}\right)^{(1-\alpha)\varepsilon_b + \alpha}.$$

Then the Engel curve for the agricultural aggregate is:

$$s_j^a \equiv \frac{P_j^a C_j^a}{P_j C_j} = \beta_a \left(\frac{P_j^a}{P_j}\right)^{1-\alpha} \left(\frac{C_j}{N_j}\right)^{(1-\alpha)(\varepsilon_a - 1)}. \tag{6}$$

Equation (6) is central to the paper. Writing that equation for both goods and taking the ratio, the price index P_j cancels out, and the relative expenditure share s_j^a / s_j^o varies with real consumption per capita with elasticity $(1-\alpha)(\varepsilon_a - \varepsilon_o)$. When $\alpha < 1$ and $\varepsilon_a < \varepsilon_o$, the agricultural expenditure share therefore declines with real consumption per capita, consistent with Engel's

law. Climate shocks that reduce real income thus raise the relative importance of food in household expenditure. This is the demand-side amplification channel of the model and Section 7 quantifies its contribution to welfare changes.

The average cost of real consumption at the country level, P_j, is given by:

$$P_j = \left[\sum_{b \in \{o,a\}} \beta_b \left(P_j^b \right)^{1-\alpha} \left(\frac{C_j}{N_j} \right)^{(1-\alpha)(\varepsilon_b - 1)} \right]^{\frac{1}{1-\alpha}}. \tag{7}$$

The homothetic CES specification corresponds to the special case of equation (7) in which $\varepsilon_b = 1$ for all $b \in \{o, a\}$. As in Comin et al. (2021), P_j can also be expressed in terms of observables and demand parameters (proof in Appendix B).

Stage 2: Demand across crops. The agricultural aggregate is a CES aggregator:

$$C_j^a = \left[\sum_k (\beta_{jk})^{\frac{1}{\kappa}} (C_{jk})^{\frac{\kappa-1}{\kappa}} \right]^{\frac{\kappa}{\kappa-1}}, \tag{8}$$

where C_{jk} is consumption of crop k in country j, $\beta_{jk} \geq 0$ is the relative weight of crop k, and $\kappa > 0$ is the elasticity of substitution across crops. Optimal demand for each crop is:

$$C_{jk} = \beta_{jk} P_{jk}^{-\kappa} \left(P_j^a \right)^{\kappa} C_j^a, \tag{9}$$

where P_{jk} is the CES price index for crop k and $P_j^a = \left[\sum_k \beta_{jk} P_{jk}^{1-\kappa} \right]^{\frac{1}{1-\kappa}}$.

Stage 3: Demand for varieties. Consumption of each crop C_{jk} depends on the consumption of varieties from different origins C_{ijk}, aggregated as:

$$C_{jk} = \left[\sum_i (\beta_{ijk})^{\frac{1}{\sigma}} (C_{ijk})^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \tag{10}$$

where σ is the substitution elasticity between varieties across all crops (Costinot et al., 2016), and $\beta_{ijk} > 0$ is the relative weight of varieties. Optimal consumption of a variety is:

$$C_{ijk} = \beta_{ijk} \; p_{ijk}^{-\sigma} \; P_{jk}^{\sigma} \; C_{jk}, \tag{11}$$

where p_{ijk} is the price of variety (i, k) in country j and $P_{jk} = \left[\sum_i \beta_{ijk} \, p_{ijk}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$.

Demand equilibrium. Combining equations (6), (9), and (11), the bilateral demand for variety (i, k) in country j can be expressed as (derivation in Appendix C):

$$p_{ijk} C_{ijk} = \beta_a \left(\frac{\beta_{ijk} \, p_{ijk}^{1-\sigma}}{P_{jk}^{1-\sigma}} \right) \left(\frac{\beta_{jk} P_{jk}^{1-\kappa}}{(P_j^a)^{1-\kappa}} \right) \left(\frac{P_j^a}{P_j} \right)^{1-\alpha} \left(\frac{C_j}{N_j} \right)^{(1-\alpha)(\varepsilon_a - 1)} P_j C_j. \tag{12}$$

This expression has the same nested CES structure as in Costinot et al. (2016). The main difference is that bilateral demand, and hence trade flows, also vary with real income per capita through the non-homotheticity parameter ε_a.

4.2. Supply

As in Costinot et al. (2016), countries differ in their agro-ecological conditions, and within each country land is composed of heterogeneous fields. Climate change affects the supply side through crop- and field-specific productivity shocks.

Technology. The non-agricultural good is produced using labor only under constant returns to scale. Let A_i^o denote labor productivity in the non-agricultural sector. Since the outside good is freely traded and used as the numeraire, profit maximization implies

$$w_i = A_i^o, \tag{13}$$

where w_i denotes the wage. Crop production uses land and labor. Land in country i is divided into fields $f \in \mathcal{F}_i$, each with area s_i^f. Each field contains a continuum of parcels indexed by $\omega \in [0,1]$. If parcel ω in field f is allocated to crop k, the output is given by:

$$Q_i^{fk}(\omega) = A_i^{fk}(\omega) \min\left\{ L_i^{fk}(\omega), \frac{N_i^{fk}(\omega)}{\nu_i^f(\omega)} \right\}, \tag{14}$$

where $L_i^{fk}(\omega)$ denotes the amount of land used on parcel ω, $N_i^{fk}(\omega)$ denotes labor, $A_i^{fk}(\omega)$ is parcel-level productivity, and $\nu_i^f(\omega)$ captures labor intensity. Land and labor are therefore perfect complements in crop production.

Parcel-level productivity and labor intensity are unobserved and drawn independently from Fréchet distributions with shape parameter $\theta > 1$. The scale parameters are chosen so that

$$A_i^{fk} = E\left[A_i^{fk}(\omega)\right], \qquad \nu_i = E\left[\nu_i^f(\omega)\right]. \tag{15}$$

The GAEZ data provide the average productivity of crop k in field f, A_i^{fk}. The parameter θ governs the extent of within-field heterogeneity.

Land allocation. Producers allocate each parcel to the crop that delivers the highest value of the marginal product, provided that this value exceeds the labor cost of operating the parcel. Since the wage is $w_i = A_i^o$, the country-specific labor-cost term can be summarized by

$$a_i \equiv A_i^o \nu_i. \tag{16}$$

The term a_i captures cross-country differences in agricultural labor costs, arising from differences in wages, labor requirements, or both.

For a given parcel ω, the producer compares the value of producing each crop,

$$p_i^1 A_i^{f1}(\omega), \ldots, p_i^K A_i^{fK}(\omega),$$

with the labor-cost threshold required to operate the parcel,

$$A_i^o \nu_i^f(\omega).$$

This labor-cost threshold is not crop-specific and therefore enters the allocation problem only once. A parcel is allocated to crop k if

$$p_i^k A_i^{fk}(\omega) = \max\left\{A_i^o \nu_i^f(\omega), p_i^1 A_i^{f1}(\omega), \ldots, p_i^K A_i^{fK}(\omega)\right\}.$$

Let π_i^{fk} denote the share of field f in country i allocated to crop k. Under the Fréchet assumption, and as in Costinot et al. (2016), this share is (see Appendices D and E):

$$\pi_i^{fk} = \frac{\left(p_i^k A_i^{fk}\right)^\theta}{a_i^\theta + \sum_{\ell \in \mathcal{K}} \left(p_i^\ell A_i^{f\ell}\right)^\theta}. \tag{17}$$

Equation (17) shows that the share of land allocated to a crop increases with its producer price p_i^k and field-level productivity A_i^{fk}. By contrast, a higher a_i reduces the share of land that is profitably allocated to crop production. The parameter θ is the elasticity of land allocation with respect to crop prices and average productivity. A lower value of θ implies greater productivity dispersion across parcels within a field, so that land allocation responds less to changes in relative prices and average productivity, including those induced by climate change.

Crop supply. Total output of crop k in country i is obtained by aggregating production over all fields $Q_i^k = \sum_f \int_0^1 Q_i^{fk}(\omega) d\omega$. As demonstrated in Appendix F, aggregate supply of crop k in country i is:

$$Q_i^k = \sum_{f \in \mathcal{F}_i} s_i^f A_i^{fk} \left[\frac{\left(p_i^k A_i^{fk}\right)^\theta}{a_i^\theta + \sum_{\ell \in \mathcal{K}} \left(p_i^\ell A_i^{f\ell}\right)^\theta}\right]^{\frac{\theta-1}{\theta}}. \tag{18}$$

Equation (18) maps field-level productivity data into country-level crop supply. Climate change enters through a shift in field-crop productivity. This shock changes land allocation, crop output, producer prices, trade flows, and consumer prices.

4.3. Trade costs and market clearing

I assume that shipping crops across countries involves iceberg trade costs $\tau_{ijk} \geq 1$, with $\tau_{iik} = 1$. Trade may also be taxed: the importer levies an ad valorem tariff $t_{ijk} \geq 0$ and the exporter an ad valorem export tax $x_{ik} \geq 0$, both equal to zero for domestic sales. The price paid in country

j for crop k produced in country i is therefore

$$p_{ijk} = (1 + t_{ijk})\,(1 + x_{ik})\,\tau_{ijk}\,p_i^k.$$

Trade costs and taxes raise the price paid by importers and shift their demand toward other suppliers. Tax revenues, T_j, are rebated lump-sum to households. The price index for crop k in country j is the CES index defined in Stage 3, $P_{jk} = \left[\sum_i \beta_{ijk}\, p_{ijk}^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$.

For each crop variety, market clearing requires that total production of crop k in country i equals total shipments to all destination countries, including iceberg trade costs:

$$Q_i^k = \sum_j \tau_{ijk} C_{ijk}. \tag{19}$$

Land allocation is defined at the field level. Each parcel is either allocated to one crop or left uncultivated:

$$\pi_i^{f\emptyset} + \sum_{k \in \mathcal{K}} \pi_i^{fk} = 1\ , \tag{20}$$

where π_i^{fk} is the share of field f allocated to crop k, and $\pi_i^{f\emptyset}$ is the share of field f that is not cultivated.

Labor is mobile across sectors within a country. Labor-market clearing requires total labor to be allocated between agriculture and the non-agricultural sector:

$$N_i = N_i^o + N_i^a\ , \tag{21}$$

where N_i^o is employment in the non-agricultural sector and $N_i^a = \sum_f \sum_k \int_0^1 N_i^{fk}(\omega)\, d\omega$ is agricultural employment.

4.4. Competitive equilibrium

An equilibrium is a vector $\{C_{ijk},\ Q_i^k,\ p_i^k,\ p_{ijk},\ \pi_i^{fk},\ P_{jk},\ P_j^a,\ P_j\}$ such that households maximize utility, producers maximize land rents, goods markets clear, and land and labor allocation conditions are satisfied. The equilibrium satisfies the demand conditions (7), (9), (11), and (12); the supply conditions (17) and (18); and the market-clearing conditions (19)–(21), given structural parameters $\{\sigma, \kappa, \alpha, \varepsilon_a, \varepsilon_o, \beta_a, \beta_o, \theta, a_i\}$, trade policy instruments $\{t_{ijk}, x_{ik}\}$, and endowments $\{L_i, N_i\}$.

The model delivers a set of equilibrium conditions that map field-level productivity shocks into changes in trade flows, prices, and household welfare. Section 5 describes the data used to discipline these conditions, including bilateral trade flows, tariffs, trade costs, expenditure shares, and field-level agricultural productivity. Section 6 then details the estimation and calibration of the structural parameters $\{\sigma, \kappa, \alpha, \varepsilon_a, \varepsilon_o, \beta_a, \beta_o, \theta, a_i\}$ that govern the demand system and land allocation. Together, these inputs make it possible to simulate the counterfactual equilibrium

under climate change and to compute the welfare decomposition in Section 7.

5. Data

This section describes the agricultural, trade, and climate data used in the quantitative analysis. I first present the field-level productivity data and the climate scenarios that define the productivity shock. I then describe the data on crop output, prices, and bilateral trade flows, and how they discipline the baseline equilibrium. Finally, I introduce the household expenditure data used to measure agricultural expenditure shares. The quantitative analysis builds on the data of Costinot et al. (2016), which I complement with agricultural expenditure shares from GTAP.

5.1. Agricultural Productivity

The counterfactual analysis compares two economies: a baseline economy without climate-induced productivity shocks and a counterfactual economy incorporating these shocks. Both economies share the same baseline data and differ only in field-level productivity, so that the counterfactual imposes the climate projected for 2071–2100 on the economy of 2010. Productivity data come from the Global Agro-Ecological Zones project (FAO et al., 2012), a joint initiative of the Food and Agriculture Organization (FAO) and the International Institute for Applied Systems Analysis (IIASA). GAEZ data combine agronomic crop modeling with detailed geospatial information, including soil type, terrain, elevation, and climate-related variables, to estimate potential yields for individual crops at a 5-arcminute resolution. The 50 countries in the sample contain 1,683,193 such grid cells, each of which corresponds to a field in the model.

GAEZ provides potential yields under the current climate and under future climate scenarios. I use yields under high inputs and rain-fed conditions. I use these data to construct two vectors of crop-level productivity at the field level: one under the current climate and one under the future climate scenario. The difference between these two vectors measures the climate-induced productivity shock that I feed into the model.

5.2. Projection Data

I complement the baseline data with projections of potential yields for 2071–2100 from GAEZ (FAO et al., 2012), simulated with the Hadley CM3 climate model (Gordon et al., 2000). Hadley CM3 is a coupled atmosphere-ocean general circulation model developed at the UK Met Office Hadley Centre, with an atmospheric resolution of 2.5° of latitude by 3.75° of longitude, and was among the main models used in the IPCC Third and Fourth Assessment Reports. It is also the climate model used by Costinot et al. (2016), which allows a direct comparison with their results. I use projections based on the IPCC Special Report on Emissions Scenarios (SRES). Four scenarios are considered: A1FI, A2, B2, and B1. Each scenario corresponds to a storyline about demographic, economic, and technological change. The A1FI scenario describes a world of rapid economic growth and rapid introduction of new technologies, with a fossil-intensive energy system and a global population that peaks in mid-century and declines thereafter. The A2

scenario describes a heterogeneous world in which economic development is regionally oriented, technological change is slower, and population grows continuously. The B2 scenario describes a world that emphasizes local solutions to economic, social, and environmental sustainability, with intermediate economic development and population growing at a lower rate than in A2. The B1 scenario describes a convergent world with the same population path as A1FI, a rapid shift toward service and information economies, and the introduction of clean and resource-efficient technologies. The best estimates of global warming by the end of the century under the SRES scenarios are 4.0°C for A1FI, 3.4°C for A2, 2.4°C for B2, and 1.8°C for B1 (Lawn, 2016). A1FI is therefore a high-emission pathway, A2 and B2 intermediate pathways, and B1 a lower-emission pathway. As in Costinot et al. (2016), A1FI serves as the benchmark scenario. In all scenarios, potential yields account for CO_2 fertilization and are computed under high-input and rain-fed conditions. Projections are available at the same grid-cell level as the baseline yields, allowing them to map directly into the fields of the model without further aggregation.

5.3. Output, Prices, and Trade Flows

Crop output and producer prices are sourced from FAOSTAT. Bilateral trade values for agricultural products are also drawn from FAOSTAT, which aggregates them from UN COMTRADE to its own product nomenclature, ensuring consistency between trade and production data at the crop level. The value of crop final consumption is computed as production plus imports minus exports, when positive. Bilateral trade costs are not observed directly: as in Costinot et al. (2016), they are absorbed into the variety weights β_{ijk}, which are calibrated to match observed bilateral trade flows. GDP and income-group classifications are drawn from the World Bank's World Development Indicators.

The final sample covers 50 countries and 10 crops selected on the basis of their share of global crop production value; the included countries account for 89.7% of world crop output value, while the selected crops represent 71.1% of global crop production. The full list of countries and crops is provided in Appendix G.

5.4. Expenditure Shares

Expenditure shares are drawn from the GTAP 9 Data Base (Aguiar et al., 2016), which provides consistent social accounting matrices for 140 regions and 57 sectors with base year 2011. For each country, I compute the share of household consumption spent on agricultural goods, defined as primary crops and livestock products, including processed rice and meat. Both domestic and imported purchases are included and valued at purchasers' prices.

Finally, I use the World Input-Output Database (Timmer et al., 2015), which provides harmonized input-output tables for 43 countries and 56 sectors over 2000–2014, to document the stylized facts.

Table 2 summarizes the climate-induced productivity shocks and agricultural expenditure shares by income group. Low- and lower-middle-income countries face the largest shocks, with average yield losses of about 34%, and the highest agricultural expenditure shares, about 26% against

Table 2: Climate-Induced Productivity Shocks and Agricultural Expenditure Shares by Income Group

Income group	Yield change (%)	ΔA_i (% of GDP)	s_j^a (%)
Low income	−34.0	−11.79	26.7
Lower-middle income	−34.2	−2.74	25.1
Upper-middle income	−33.0	−1.10	10.6
High income	−0.6	−0.01	3.1
All countries	−25.1	−2.41	14.9

Notes: For each country, yield change is the average change in field-level productivity between the baseline and future climates, weighted by observed crop output, and ΔA_i is the corresponding productivity change expressed in percent of GDP. s_j^a is the share of household consumption spent on agricultural goods. Rows report unweighted averages of country values within each income group. *Source:* GAEZ, FAOSTAT, and GTAP data.

3.1% in high-income countries. Country-level values are reported in Table G.3.

6. Estimation

In this section, I describe how the model is brought to the data. The empirical implementation follows the demand and supply blocks developed in Section 4. I first construct the baseline economy using data on production, land use, bilateral trade, expenditure, income, population, and field-crop productivity. I then estimate the demand-side parameters using the non-homothetic expenditure-share equations from Section 4.1 and calibrate the supply side using the land-allocation and crop-supply equations from Section 4.2. Finally, I solve the model under baseline and counterfactual productivity conditions and compute welfare changes.

6.1. Demand-side estimation

The demand-side estimation relies on the non-homothetic equations derived in Section 4.1. In particular, I use the agricultural expenditure share in equation (6), which links the share of expenditure allocated to agriculture to relative prices and consumption per capita. For each country i, the agricultural expenditure share is

$$s_i^a = \beta_a \left(\frac{P_i^a}{P_i}\right)^{1-\alpha} \left(\frac{C_i}{N_i}\right)^{(1-\alpha)(\varepsilon_a-1)}, \tag{22}$$

in the same way that the non-agricultural expenditure share is

$$s_i^o = \beta_o \left(\frac{P_i^o}{P_i}\right)^{1-\alpha} \left(\frac{C_i}{N_i}\right)^{(1-\alpha)(\varepsilon_o-1)}. \tag{23}$$

Taking the ratio of equations (22) and (23) and normalizing $\varepsilon_o = 1$ gives the estimating equation

$$\frac{s_i^a}{s_i^o} = \frac{\beta_a}{\beta_o} \left(\frac{P_i^a}{P_i^o} \right)^{1-\alpha} \left(\frac{C_i}{N_i} \right)^{(1-\alpha)(\varepsilon_a - 1)}. \tag{24}$$

I estimate the upper-level demand parameters from equation (24) by nonlinear least squares, for a given value of α, minimizing the squared distance between the relative expenditure share observed in the data and the relative expenditure share implied by the model:

$$\min_{\beta_a, \beta_o, \varepsilon_a} \sum_i \left[\frac{\beta_a}{\beta_o} \left(\frac{P_i^a}{P_i^o} \right)^{1-\alpha} \left(\frac{C_i}{N_i} \right)^{(1-\alpha)(\varepsilon_a - 1)} - \frac{s_i^a}{s_i^o} \right]^2, \tag{25}$$

with $\beta_a + \beta_o = 1$, $\beta_a \geq 0$ and $\beta_o \geq 0$. The relative price term and consumption per capita are constructed from the baseline data; consumption per capita C_i/N_i is measured as total expenditure divided by employment, E_i/N_i. The results are reported in Table 3.

6.2. Supply-side estimation

The supply side estimation relies on the land-allocation equation (17) and the crop-supply equation (18) of section 4.2. Following Costinot et al. (2016), parcel-level productivity and labor requirements are unobserved and drawn from Fréchet distributions with a common shape parameter θ. In the baseline calibration, I take θ from Costinot et al. (2016). The country-specific labor-cost shifter a_i, defined in equation (16), is calibrated to match observed cultivated land in each country. Specifically, for a given value of θ, the model share of field f allocated to crop k is

$$\pi_{ifk}(\theta, a_i) = \frac{\left(p_i^k A_i^{fk} \right)^\theta}{a_i^\theta + \sum_{\ell \in \mathcal{K}} \left(p_i^\ell A_i^{f\ell} \right)^\theta}. \tag{26}$$

From this share, the predicted cultivated area in country i is:

$$L_i(\theta, a_i) = \sum_{k \in \mathcal{K}} \sum_{f \in \mathcal{F}_i} s_i^f \; \pi_{ifk}(\theta, a_i). \tag{27}$$

Following Costinot et al. (2016), a_i is calibrated, conditional on θ, so that the predicted cultivated area exactly matches its observed counterpart:

$$L_i(\theta, a_i) = L_i \quad \forall i. \tag{28}$$

The country-specific labor-cost shifter a_i governs the profitability threshold for land use. A higher value raises the labor-cost threshold required to operate parcels and reduces the share of land that is profitably cultivated, consistent with equation (20). a_i is obtained from equation (29) by solving a nonlinear calibration problem:

$$\min_{a_i>0} \left[\log L_i(a_i;\theta) - \log L_i^{\text{data}}\right]^2. \tag{29}$$

Given A_i^{fk}, a_i, θ, and producer prices p_i^k, aggregate crop supply q_i^k is computed using equation (18).

6.3. Climate counterfactuals

The counterfactual analysis compares a baseline economy with a counterfactual economy incorporating climate-induced productivity shocks. These shocks are derived from GAEZ potential yields, vary across fields and crops, and enter the model through the productivity term A_i^{fk} in the land-allocation and crop-supply equations. Let A_i^{fk} denote the potential yield of crop k in field f of country i under current climate conditions, and $A_i^{fk\prime}$ the corresponding yield under the climate projected for 2071–2100 under a given SRES scenario. The field-crop productivity shock is defined as

$$\Delta A_i^{fk} = A_i^{fk\prime} - A_i^{fk}. \tag{30}$$

In the counterfactual equilibrium, equation (18) becomes

$$Q_i^{k\prime} = \sum_{f\in\mathcal{F}_i} s_i^f A_i^{fk\prime} \left[\frac{\left(p_i^{k\prime} A_i^{fk\prime}\right)^\theta}{a_i^\theta + \sum_{\ell\in\mathcal{K}} \left(p_i^{\ell\prime} A_i^{f\ell\prime}\right)^\theta}\right]^{\frac{\theta-1}{\theta}}. \tag{31}$$

All structural parameters are held fixed across the baseline and counterfactual equilibria. Thus, differences between the two equilibria are driven by the productivity shift from A_i^{fk} to $A_i^{fk\prime}$. These shocks affect land allocation, crop output, producer prices, delivered prices, bilateral trade flows, price indices, and welfare.

6.4. Welfare Measurement

To quantify the consequences of climate change, I measure household welfare using the equivalent variation (EV) in fixed final preferences (Baqaee and Burstein, 2023; Baqaee et al., 2024; Jaravel and Lashkari, 2024). The EV evaluates the income gain or loss associated with moving from the baseline to the counterfactual equilibrium, holding utility fixed at the post-shock level u':

$$\Delta W = \frac{e(P,\, u') - e(P,\, u)}{e(P,\, u)}, \tag{32}$$

where $e(\cdot)$ is the expenditure function derived from the non-homothetic CES preferences, u and u' denote utility in the baseline and counterfactual equilibria, $P = (P_j^a, P_j^o)$ denotes the vector of agricultural and non-agricultural prices at the baseline equilibrium, and $P' = (P_j^{a\prime}, P_j^{o\prime})$ denotes the corresponding prices at the counterfactual equilibrium induced by the climate productivity

shock. [3] Hence, ΔW measures, at baseline prices P, the proportional change in income that would leave households at the post-shock utility level u'. Equivalently, it equals the change in income minus the additional cost of reaching u' at counterfactual prices P' rather than at baseline prices P. A negative value indicates a welfare loss, and a positive value a welfare gain.

As shown in Appendix H, the total welfare effect can be decomposed into three components:

$$\Delta W = \underbrace{\frac{e(P',\, u')}{e(P,\, u)}}_{\text{income effect}} \times \underbrace{\frac{e(P,\, u)}{e(P',\, u)}}_{\text{price effect}} \times \underbrace{\frac{e(P,\, u')/e(P',\, u')}{e(P,\, u)/e(P',\, u)}}_{\text{structural change effect}} - 1. \tag{33}$$

The income effect captures the change in nominal income, which includes agricultural income and trade-tax revenue. The price effect captures welfare losses arising from climate-induced changes in relative prices, evaluated at baseline utility. The structural change effect captures the additional welfare impact operating through income-driven shifts in expenditure shares: it compares the cost-of-living index at the post-shock and baseline utility levels, and equals one under homothetic preferences. The decomposition in equation (33) isolates these channels and is a key object of interest.

7. Results

7.1. Parameter values

Table 3 reports the parameter values used in the simulations. The substitution elasticities κ and σ, and the heterogeneity parameter within fields θ, are taken from Costinot et al. (2016). The upper-level demand parameters $\{\beta_a, \beta_o, \varepsilon_a\}$ are estimated using equation (25), with $\alpha = 0.5$ and $\varepsilon_o = 1$. The country-specific labor-cost shifter a_i is calibrated separately to match observed cultivated land using equation (29). I obtain a point estimate of $\varepsilon_a = 0.083$ (robust s.e. 0.241), reported in Table 3. The estimate is significantly below one and consistent with the conditions specified in Section 4.1, namely $\alpha > 0$ and $0 < \varepsilon_a < 1$. This implies a strong departure from homotheticity in food demand.

The choice of $\alpha = 0.5$ is consistent with estimates reported in the literature on structural transformation. Comin et al. (2021) estimate non-homothetic CES preferences using household-level data from the Consumer Expenditure Survey and cross-country data. The value $\alpha = 0.5$ lies within the range of their cross-country estimates, between 0.25 and 0.63, and matches their estimate for the world sample with trade controls. It is above their household-level estimates (0.20 to 0.33) and the value estimated by Nath (2025) (0.27), but below the estimate of 0.85 reported by Herrendorf et al. (2013) for final consumption expenditure. Since $\alpha < 1$, agricultural and non-agricultural goods are gross complements. Holding utility constant, a one percent increase in the relative price of agricultural goods increases the ratio of agricultural to non-agricultural expenditure shares by approximately $(1 - \alpha) = 0.5$ percent.

[3] As in Comin et al. (2021), the expenditure function derived from the non-homothetic CES preferences is $e(\beta, P, u) = \left[\sum_b \beta_b (P_j^b)^{1-\alpha}\, u^{(1-\alpha)\varepsilon_b}\right]^{1/(1-\alpha)}$.

Table 3: Parameter Values

Parameter	Description	Value
Upper-level demand		
β_a	Preference weight, agricultural goods	0.62 (0.14)
β_o	Preference weight, non-agricultural goods	0.38 (0.14)
ε_a	Non-homotheticity, agricultural goods	0.083 (0.24)
α	Substitution, agricultural vs. other goods	0.50
ε_o	Non-homotheticity, non-agricultural goods	1.00
Lower-level demand		
κ	Substitution between crops	2.82
σ	Substitution between origins of a crop	5.40
Supply		
θ	Productivity heterogeneity within fields	2.46

Notes: β_a, β_o and ε_a are estimated by nonlinear least squares, with $\beta_a + \beta_o = 1$; robust standard errors in parentheses. The 95% confidence interval of ε_a is $[-0.40, 0.57]$. α is fixed and ε_o is normalized to one. κ, σ and θ are taken from existing estimates. *Source:* Author's calculations based on GTAP and WDI data.

Second, the estimated non-homotheticity parameter $\varepsilon_a = 0.083 < 1$ implies that the agricultural expenditure share declines with real consumption, consistent with Engel's law (Herrendorf et al., 2014; Matsuyama, 2019). This estimate lies within the range of the cross-country estimates of Comin et al. (2021). The implied income elasticity of agricultural demand, about 0.55, is also consistent with the values of 0.37 to 0.56 reported by Comin et al. across their household- and country-level applications, and close to the estimate of 0.48 reported by Nath (2025). Holding relative prices constant, a one percent increase in the model's real consumption index reduces the ratio of agricultural to non-agricultural expenditure shares by approximately $(1-\alpha)(\varepsilon_a - 1) = -0.46$ percent. Conversely, climate-induced reductions in real consumption raise food expenditure shares, increasing exposure to food price shocks, particularly in countries where food already accounts for a large share of consumption expenditure.

Finally, I assess the robustness of the demand estimates to the calibrated value of α. Appendix I reports estimates of ε_a for $\alpha \in [0.2, 0.5]$. Across this range, ε_a varies between 0.083 and 0.351, while $(1-\alpha)(\varepsilon_a - 1)$ remains between -0.52 and -0.46. These results suggest that the income effect underlying the welfare analysis is robust to the choice of α. Regardless of the assumed degree of complementarity between agricultural and non-agricultural goods, higher real consumption is associated with a lower agricultural expenditure share, holding relative prices constant. This mechanism helps explain why climate-induced agricultural productivity shocks can have larger welfare consequences in lower-income economies.

7.2. Climate shocks and welfare

Table 4 reports the welfare effects of climate change by country, while Table 5 summarizes the results by income group. At the aggregate level, climate change reduces world welfare by 0.83% of income under non-homothetic preferences, compared with 0.74% under homothetic prefer-

ences. With non-homothetic preferences, declining real income increases the share of household expenditure devoted to food, precisely when food becomes more expensive. Food demand also falls less as income declines, so that agricultural prices rise more than under homothetic preferences. Section 7.3 quantifies these channels. Weighted by employment rather than GDP, the average welfare loss reaches 3.77%, reflecting the larger losses experienced by lower-income economies, which account for a large share of world employment. These losses are unevenly distributed: lower-middle-income countries lose 8.53% of income and low-income countries 6.23%, compared with 1.09% in upper-middle-income countries and 0.04% in high-income countries. Non-homothetic preferences raise the losses of every income group, and most those of lower-middle-income countries, by 1.02 percentage points (Table 5).

Table 4: Welfare Effects of Climate Change by Country (% of Income)

Country	ΔA_i (% GDP)	Homo-thetic	Non-homo-thetic	Country	ΔA_i (% GDP)	Homo-thetic	Non-homo-thetic
Low income				Upper-middle income			
Congo, Dem. Rep.	−7.04	−10.83	−11.48	Algeria	−0.88	−5.00	−5.38
Ethiopia	−3.75	−2.90	−3.26	Argentina	0.20	0.72	0.74
Malawi	−38.41	−16.80	−16.09	Brazil	−1.94	−4.48	−4.63
Tanzania	−5.91	−8.95	−9.64	China	−0.46	0.24	0.24
Uganda	−3.82	−1.08	−0.93	Colombia	−1.47	−5.73	−5.97
				Ecuador	−4.61	−1.73	−1.85
Lower-middle income				Iran	−0.94	−2.49	−2.63
Bangladesh	−3.44	−16.71	−18.86	Kazakhstan	−0.49	−4.82	−5.07
Burma	−5.96	−7.94	−8.03	Malaysia	−0.35	−1.51	−1.57
Côte d'Ivoire	−2.52	−13.44	−15.18	Mexico	−0.39	−1.22	−1.25
Cameroon	−3.07	−10.72	−11.79	Romania	−0.60	−4.69	−5.02
Egypt	−1.96	−0.88	−0.98	Russia	−0.03	7.21	7.07
Ghana	−6.37	−34.31	−42.58	South Africa	−0.54	−3.30	−3.43
India	−1.84	−5.28	−5.63	Thailand	−2.61	−11.26	−11.94
Indonesia	−1.26	−5.20	−5.43	Turkey	−0.30	0.07	0.06
Morocco	−1.57	−9.33	−10.15	Venezuela	−2.24	−14.93	−16.26
Nigeria	−2.59	−35.18	−44.21				
Pakistan	−1.74	−3.23	−3.50	High income			
Philippines	−1.32	−4.00	−4.15	Australia	−0.23	−0.32	−0.32
Sudan	−3.35	−37.42	−45.98	Canada	0.26	0.76	0.77
Ukraine	−0.54	−0.53	−0.51	France	−0.08	−0.16	−0.17
Uzbekistan	−2.22	1.51	1.50	Germany	0.01	−0.05	−0.05
Viet Nam	−4.16	−4.68	−4.77	Greece	−0.13	0.13	0.13
				Italy	−0.07	−0.06	−0.07
				Japan	0.04	0.07	0.06
				Korea, Rep.	0.17	−1.01	−1.04
				Netherlands	0.08	0.41	0.40
				Poland	0.06	0.96	0.95
				Spain	−0.09	0.35	0.34
				United Kingdom	0.00	0.76	0.76
				United States	−0.14	−0.26	−0.27
World (GDP-weighted)					−0.34	−0.74	−0.83

Notes: Equivalent variation in percent of baseline income under the Hadley CM3 A1FI scenario. ΔA_i is the climate-induced productivity shock in percent of GDP. The homothetic case sets $\varepsilon_a = \varepsilon_o = 1$; the non-homothetic case uses $\varepsilon_a = 0.083$. Income groups follow the World Bank classification. *Source:* Author's calculations using GAEZ, FAOSTAT, GTAP and WDI data.

At the country level, the largest losses occur in Sudan (46.0%), Nigeria (44.2%), and Ghana (42.6%), followed by Bangladesh, Venezuela, and Malawi. In Sudan, Nigeria, and Ghana, the

welfare loss is several times larger than the productivity shock expressed in percent of GDP. Conversely, Malawi, which faces the largest productivity shock (38.4% of GDP), loses less (16.1%). A few countries gain, among them Russia (7.1%), Uzbekistan, Poland, Canada, and the United Kingdom, although the productivity shock is slightly negative in Russia. Section 7.3 helps explain these differences. Accounting for non-homothetic preferences amplifies welfare losses particularly in several lower-middle-income countries: from 37.4% to 46.0% in Sudan, from 35.2% to 44.2% in Nigeria, and from 34.3% to 42.6% in Ghana, while the welfare changes of high-income countries are almost unaffected.

Table 5: Welfare Effects of Climate Change by Income Group (% of Income)

Income group	Homothetic	Non-homothetic	Difference
Low income	−5.90	−6.23	−0.33
Lower-middle income	−7.51	−8.53	−1.02
Upper-middle income	−0.97	−1.09	−0.12
High income	−0.03	−0.04	−0.01
World (GDP-weighted)	−0.74	−0.83	−0.09
World (employment-weighted)	−3.35	−3.77	−0.42

Notes: Equivalent variation in percent of baseline income under the Hadley CM3 A1FI scenario. Income-group averages are GDP-weighted; world averages are weighted by GDP or employment, as indicated. The homothetic case sets $\varepsilon_a = \varepsilon_o = 1$; the non-homothetic case uses $\varepsilon_a = 0.083$. Income groups follow the World Bank classification. *Source:* Author's calculations based on GAEZ, FAOSTAT, GTAP and WDI data.

For comparison, Costinot et al. (2016) find a global welfare loss of 0.26% of GDP under quasi-linear preferences, in which expenditure on crops is fixed. Under homothetic preferences, the model yields a larger loss (0.74%), reflecting differences in the demand calibration, particularly the use of broader GTAP agricultural expenditure shares rather than expenditure on the ten crops alone.

7.3. Welfare Decomposition

Table 6 decomposes the welfare effects into the price, structural change, and income effects, as defined in equation (33). Climate shocks raise the agricultural price index by 13.4% at the world level (12.7% under homothetic preferences), by about 41% in low-income and 46% in lower-middle-income countries, and by 5.5% in high-income countries.

The price effect accounts for most of the welfare loss in every income group. In lower-middle-income countries, it amounts to 8.10% of income, against 0.85% for the structural change effect. At the aggregate level, the structural change effect represents about 10% of the total loss. The structural change effect is zero by construction under homothetic preferences. Non-homotheticity also raises the price effect, because food demand falls less as income declines. The income effect is positive in all income groups and largest in low-income countries (1.99%), where higher agricultural prices raise farm income. It offsets only a small part of the price effect.

The decomposition also explains why welfare losses are not proportional to the productivity shocks: what matters is how the shock is transmitted through prices. In Sudan, Nigeria, and

Table 6: Decomposition of the Welfare Effects of Climate Change (% of Income)

Income group	Agricultural price (%)		Price	Structural	Income	Total
	Homothetic	Non-homothetic	effect	change effect	effect	ΔW
Low income	37.8	40.9	−7.87	−0.35	1.99	−6.23
Lower-middle income	42.4	45.6	−8.10	−0.85	0.42	−8.53
Upper-middle income	27.2	28.1	−1.05	−0.10	0.06	−1.09
High income	5.1	5.5	−0.09	0.00	0.05	−0.04
World	12.7	13.4	−0.83	−0.08	0.08	−0.83

Notes: Hadley CM3 A1FI scenario. Agricultural price is the percent change in the consumer price index of agricultural goods under homothetic ($\varepsilon_a = \varepsilon_o = 1$) and non-homothetic ($\varepsilon_a = 0.083$) preferences. The welfare decomposition is reported for non-homothetic preferences, in percent of baseline income. GDP-weighted averages within each income group; income groups follow the World Bank classification. *Source:* Author's calculations based on GAEZ, FAOSTAT, GTAP and WDI data.

Ghana, the consumer price of agricultural goods more than doubles and food absorbs a large share of expenditure, so that the price effect dominates and non-homotheticity amplifies it: the structural change effect accounts for about a fifth of the total loss (Appendix Table H.1). In Malawi, which faces the largest productivity shock, agriculture accounts for a large share of income, and higher agricultural prices raise farm income, which offsets about half of the price effect. In Russia, productivity rises for wheat, maize, and tomato, the consumer price of agricultural goods falls, and consumers gain despite a slightly negative average productivity shock.

I next examine whether adaptation can mitigate welfare losses. Appendix Table H.2 presents a counterfactual in which low-income and lower-middle-income countries reduce their climate-induced yield losses by 50%. Unlike the crop and land reallocation already embedded in the model, this experiment represents costless yield improvements, such as irrigation or new varieties, and gives an upper bound on their gains. The welfare losses of these countries fall by more than half, from 6.23% to 1.76% and from 8.53% to 4.08% of income, because the rise in agricultural prices is much smaller. Adaptation in other countries adds little, which indicates that the losses of lower-income countries stem mainly from their own productivity shocks rather than from world prices. Adaptation benefits households as consumers more than as producers: by limiting the rise in food prices, it reduces the cost of living but also the gains from higher farm prices.

The implication of these results is that relative price changes matter more than sectoral reallocation. The primary welfare channel of climate change is not how much the economy restructures, but how much consumers pay for essential goods, especially in countries where food makes up a large share of consumption. Non-homothetic preferences amplify this channel but do not replace it. These effects are consistent with Hottman and Monarch (2018) and Atkin et al. (2020), who show that lower-income households tend to experience the highest price inflation.

7.4. Sensitivity Analysis

I assess the sensitivity of the results to alternative climate scenarios, to different values of the elasticity of substitution between agricultural and non-agricultural goods, and to the calibration of agricultural expenditure shares.

Table 7 reports the results under alternative SRES emissions scenarios simulated with Hadley CM3. Welfare losses increase with the severity of warming: the world loss falls from 0.83% under A1FI to 0.77% under A2, 0.43% under B2, and 0.20% under B1. Lower-middle-income countries remain the largest losers in every scenario, and non-homothetic preferences raise the world loss by a similar proportion across scenarios.

Table 7: Welfare Effects under Alternative Emissions Scenarios (% of Income)

	ΔA	World (GDP-weighted)		Income group			
SRES scenario	(% of GDP)	Non-homothetic	Homothetic	Low	Lower-middle	Upper-middle	High
A1FI	−0.34	−0.83	−0.74	−6.23	−8.53	−1.09	−0.04
A2	−0.32	−0.77	−0.68	−4.70	−8.60	−0.77	−0.05
B2	−0.20	−0.43	−0.38	−1.98	−4.66	−0.65	0.04
B1	−0.13	−0.20	−0.17	−2.39	−2.80	−0.31	0.08

Notes: Equivalent variation in percent of baseline income. All scenarios use GAEZ potential yields for 2071–2100 simulated with Hadley CM3, with CO_2 fertilization, high inputs and rain-fed conditions. ΔA is the GDP-weighted average of the country productivity shocks. The non-homothetic case uses $\varepsilon_a = 0.083$; the homothetic case sets $\varepsilon_a = \varepsilon_o = 1$. Income-group columns are GDP-weighted averages under non-homothetic preferences; income groups follow the World Bank classification. *Source:* Author's calculations based on GAEZ, FAOSTAT, GTAP and WDI data.

Table 8 reports alternative demand specifications. Appendix I re-estimates the demand parameters for lower values of the elasticity of substitution between agricultural and non-agricultural goods. Lower values of α reduce substitution away from food and raise welfare losses: with the value of Comin et al. (2021), $\alpha = 0.26$, the world loss reaches 1.00%. The benchmark $\alpha = 0.5$ therefore yields the smallest losses of the range, and the structural change effect remains stable across values of α. Assigning every country the sample-average agricultural expenditure share reduces the losses of low-income countries by more than half and raises those of high-income countries, which indicates that cross-country differences in food shares drive the distribution of the welfare effects. Finally, imputing the shares of the countries that are not separate regions in GTAP from the sample median leaves the world loss almost unchanged.

7.5. Trade Policy Counterfactuals

I consider two sets of trade policy scenarios. The first is a stylized experiment that measures the role of trade integration. Starting from a baseline equilibrium in which every country levies a 30% export tax on all crops, I compare the counterfactual equilibrium under the climate shock when the tax is maintained with the one in which it is removed. The second set starts from the observed baseline without taxes. I introduce export restrictions after climate shocks: either the three largest exporters of each crop, identified from observed trade flows (Appendix Table J.1),

Table 8: Robustness of the Welfare Effects (% of Income)

Specification	World (GDP-weighted)	Income group			
		Low	Lower-middle	Upper-middle	High
Benchmark	−0.83	−6.23	−8.53	−1.09	−0.04
Homothetic preferences	−0.74	−5.90	−7.51	−0.97	−0.03
Lower substitutability	−1.00	−6.57	−9.29	−1.50	−0.07
Uniform agricultural share	−1.46	−2.96	−5.24	−2.74	−0.68
Median imputation	−0.79	−5.54	−7.98	−1.05	−0.04

Notes: Equivalent variation in percent of baseline income under the Hadley CM3 A1FI scenario, GDP-weighted averages within each income group. Benchmark: $\alpha = 0.5$, $\varepsilon_a = 0.083$. Homothetic preferences: $\varepsilon_a = \varepsilon_o = 1$. Lower substitutability: $\alpha = 0.26$, with $\varepsilon_a = 0.318$ re-estimated. Uniform agricultural share: every country is assigned the sample-average share. Median imputation: the shares of Burma, Sudan, Uzbekistan, Algeria, and the Democratic Republic of Congo, which are not separate regions in GTAP, are set to the sample median. *Source:* Author's calculations based on GAEZ, FAOSTAT, GTAP and WDI data.

impose a 25% or 50% export tax, or exporters set export taxes so as to absorb part or all of the climate-induced increase in their domestic producer price (price insulation). In both sets, each policy is compared with climate shocks without the policy, from the same baseline.

Table 9 reports the results. Removing the export tax reduces the world welfare loss from 0.85% to 0.76% of income. Lower-middle-income economies benefit the most, whereas low-income economies lose slightly. In contrast, export restrictions amplify welfare losses. A 50% export tax imposed by the three largest exporters increases the global welfare loss, and full price insulation increases it further. Lower-middle-income economies bear the largest increase, while restricting insulation to staple crops, namely wheat, rice, and maize, which are frequent targets of trade interventions (Martin et al., 2026), leaves the world loss almost unchanged.

Appendix Table J.2 separates the effect of each policy under climate shocks from its effect without climate shocks. At the aggregate level, most of the effect of fixed export taxes would arise even without climate change: removing the 30% tax raises world welfare by 0.08% of income in the absence of the shock, so that trade integration reduces the climate-induced loss only marginally. The part of the policy effects that is specific to climate change is concentrated in lower-middle-income countries, which face the largest increase in agricultural prices. Full price insulation, which is triggered only by the rise in prices, is the most costly response for these countries. Consistent with Hsiao et al. (2026), these results show that export restrictions adopted in response to climate shocks shift part of their burden onto lower-income economies.

8. Policy implications

The results have three main implications for policy. First, climate change affects welfare mainly through agricultural prices. Welfare losses depend not only on the magnitude of productivity shocks, but also on their transmission through food prices and countries' initial food expenditure shares. Non-homothetic preferences amplify these losses, particularly in lower-income economies where food accounts for a large share of consumption. Assessments of climate vulnerability should therefore consider exposure to food price increases alongside productivity losses. The

Table 9: Trade Policy Counterfactuals

	30% export tax		No policy	Export tax, top-3		Insulation, all crops		Insulation, staples	
Income group	Maintained	Removed		25%	50%	50%	100%	50%	100%
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Low income	−6.07	−6.20	−6.23	−6.27	−6.29	−6.34	−6.37	−6.27	−6.30
Lower-middle income	−8.76	−8.39	−8.53	−8.71	−8.86	−8.72	−8.93	−8.49	−8.53
Upper-middle income	−1.17	−1.02	−1.09	−1.15	−1.20	−1.19	−1.28	−1.12	−1.16
High income	−0.02	0.03	−0.04	−0.06	−0.08	−0.06	−0.07	−0.04	−0.04
World	−0.85	−0.76	−0.83	−0.87	−0.91	−0.88	−0.93	−0.84	−0.85

Notes: Equivalent variation in percent of baseline income under the Hadley CM3 A1FI scenario, GDP-weighted averages within each income group. Columns (2)–(3) start from a baseline equilibrium with a 30% export tax on all crops, which is maintained or removed in the counterfactual; Columns (4)–(10) start from the observed baseline without taxes; columns (5)–(10) are compared with column (4). Columns (5)–(6): the three largest exporters of each crop in observed trade flows impose a 25% or 50% export tax. Columns (7)–(10): exporters (exports above 10% of sales) set export taxes so as to absorb 50% or 100% of the climate-induced rise in their domestic producer price; columns (9)–(10) restrict insulation to wheat, maize, and rice, for exporters that sell at least 20% of their output domestically. Export taxes are bounded at 500%, which amounts to a de facto export ban. *Source:* Author's calculations based on GAEZ, FAOSTAT, GTAP and WDI data.

importance of the price channel also suggests a role for policies that protect household food consumption, although their effectiveness is not directly evaluated in the model.

Second, improving agricultural productivity in lower-income countries can substantially mitigate welfare losses. In the adaptation counterfactual, reducing climate-induced yield losses by 50% in low-income and lower-middle-income countries considerably improves their welfare, primarily by limiting increases in agricultural prices. Extending adaptation to other countries generates further global gains, but relatively small additional benefits for lower-income economies. Investments in irrigation, improved crop varieties, and agricultural research could contribute to such productivity improvements, although their benefits would need to be weighed against implementation costs.

Third, trade policies can either facilitate or hinder adjustment to climate shocks. Removing export taxes generates welfare gains that are largely independent of climate change at the global level. By contrast, export restrictions and price-insulation policies amplify global welfare losses, with particularly adverse effects on lower-middle-income economies. These findings highlight the importance of maintaining access to international agricultural markets and improving trade policy coordination during periods of high food prices, consistent with the objectives of the Agricultural Market Information System (Ferguson and Ubilava, 2022).

These implications concern country-level welfare effects. Because the model features a representative household in each country, it does not capture differences between net food buyers and sellers, who may be affected differently by changes in agricultural prices. Assessing the distributional effects of targeted transfers and the costs of agricultural adaptation would be valuable extensions.

9. Conclusion

Much of the discussion of climate change in agriculture focuses on how much crop productivity declines and where. Yet the welfare consequences of these shocks also depend on how they affect food prices and on the share of household expenditure devoted to food. This paper quantifies these consequences by combining field-level potential yields, bilateral trade flows, and country-level expenditure data in a multi-country Ricardian trade model with non-homothetic preferences. I find that climate-induced productivity shocks reduce global welfare by 0.83% of income, with much larger losses in lower-income economies, particularly lower-middle-income countries.

The welfare decomposition shows that higher agricultural prices account for most of the losses across income groups. The structural change effect, which captures income-driven adjustments in expenditure shares, plays a smaller role, while higher agricultural income offsets part of the price effect. Non-homothetic preferences amplify welfare losses relative to the homothetic benchmark, particularly in lower-middle-income economies. Differences in food expenditure shares across countries also play a central role in determining the distribution of these losses. An adaptation counterfactual shows that reducing climate-induced yield losses in lower-income countries can substantially improve their welfare, mainly by limiting increases in agricultural prices.

Trade policies also influence the welfare consequences of climate change. Removing export taxes improves global welfare, although most of these gains arise independently of climate shocks. By contrast, export restrictions and price insulation by exporters amplify welfare losses, particularly in lower-middle-income economies. Together, these findings highlight the potential benefits of agricultural adaptation in vulnerable countries and the importance of avoiding trade restrictions that amplify the rise in food prices.

The framework has several limitations that suggest directions for future research. It relies on a representative household in each country and therefore does not capture differences between net food buyers and sellers. The adaptation experiment treats yield improvements as costless, while the counterfactuals apply projected climate-induced productivity changes to the baseline economy. Extending the framework to multiple sectors with imperfect labor mobility, as in Nath (2025), and incorporating household heterogeneity would allow a more detailed assessment of the distributional consequences of climate change.

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Appendix

Appendix A. Household's Optimization Problem

In each country, the representative agent derives utility from the consumption of the outside good C_j^o and the agricultural aggregate C_j^a, with preferences given by the Non-Homothetic CES utility function:

$$\sum_{b\in\{o,a\}} (\beta_b)^{\frac{1}{\alpha}} \left(\frac{C_j}{N_j}\right)^{\frac{\varepsilon_b(1-\alpha)}{\alpha}} \left(\frac{C_j^b}{N_j}\right)^{\frac{\alpha-1}{\alpha}} = 1, \tag{A.1}$$

where $U_j = E_j/(P_j N_j) = C_j/N_j$. The representative household maximizes U_j subject to the budget constraint:

$$E_j = \sum_b P_j^b C_j^b = w_j N_j + r_j L_j + T_j + D_j, \tag{A.2}$$

with $\sum_j D_j = 0$. The household's optimization problem implies that consumers select each good C_j^b, $b \in \{a, o\}$ that maximize aggregate consumption. The representative household's Lagrangian is:

$$\mathcal{L} = U_j + \rho\left[1 - \sum_b (\beta_b)^{\frac{1}{\alpha}} \left(\frac{C_j}{N_j}\right)^{\varepsilon_b\frac{(1-\alpha)}{\alpha}} \left(\frac{C_j^b}{N_j}\right)^{\frac{\alpha-1}{\alpha}}\right] + \Lambda\left[w_j N_j + r_j L_j + T_j + D_j - \sum_b P_j^b C_j^b\right]$$

with ρ and Λ the appropriate Lagrange multipliers. The first-order condition for C_j^b is:

$$0 = -\rho\,(\beta_b)^{\frac{1}{\alpha}} \left(\frac{C_j}{N_j}\right)^{\varepsilon_b\frac{(1-\alpha)}{\alpha}} \left(\frac{\alpha-1}{\alpha}\right) \left(\frac{C_j^b}{N_j}\right)^{\frac{\alpha-1}{\alpha}} \left(\frac{N_j}{C_j^b}\right) \left(\frac{1}{N_j}\right) - \Lambda\ P_j^b$$

$$\Rightarrow (\beta_b)^{\frac{1}{\alpha}} \left(\frac{C_j}{N_j}\right)^{\varepsilon_b\frac{(1-\alpha)}{\alpha}} \left(\frac{C_j^b}{N_j}\right)^{\frac{\alpha-1}{\alpha}} = \frac{\Lambda\alpha}{\rho(1-\alpha)} P_j^b\ C_j^b$$

Using constraints (A.1) and (A.2), we get:

$$\frac{\rho(1-\alpha)}{\Lambda\alpha} = \sum_b P_j^b C_j^b = E_j \tag{A.3}$$

which further implies that:

$$(\beta_b)^{\frac{1}{\alpha}} \left(\frac{C_j}{N_j} \right)^{\varepsilon_b \frac{(1-\alpha)}{\alpha}} \left(\frac{C_j^b}{N_j} \right)^{\frac{\alpha-1}{\alpha}} = \frac{P_j^b C_j^b}{E_j} \tag{A.4}$$

Therefore, the optimal consumption of good C_j^b can be written as :

$$C_j^b = \beta_b \; N_j \left(\frac{P_j^b N_j}{E_j} \right)^{-\alpha} \left(\frac{C_j}{N_j} \right)^{\varepsilon_b (1-\alpha)} \tag{A.5}$$

If we define P_j as the ideal consumer price index for the aggregate consumption of goods C_j, such that $E_j = P_j \; C_j$, then we have:

$$C_j^b = \beta_b \; N_j \left(\frac{P_j^b}{P_j} \right)^{-\alpha} \left(\frac{C_j}{N_j} \right)^{(1-\alpha)\varepsilon_b + \alpha} \tag{A.6}$$

For agri-food products, we have :

$$C_j^a = \beta_a \; N_j \left(\frac{P_j^a}{P_j} \right)^{-\alpha} \left(\frac{C_j}{N_j} \right)^{(1-\alpha)\varepsilon_a + \alpha} .$$

Multiplying both sides by $\frac{P_j^a}{P_j}$ results in an Engel equation that characterizes the share of spending allocated to agri-food products:

$$\frac{P_j^a C_j^a}{P_j} = \beta_a \left(\frac{P_j^a}{P_j} \right)^{1-\alpha} \left(\frac{C_j}{N_j} \right)^{(1-\alpha)(\varepsilon_a - 1)} (C_j) \tag{A.7}$$

$$\frac{P_j^a \; C_j^a}{P_j \; C_j} = \mathbf{s}_j^a = \beta_a \left(\frac{P_j^a}{P_j} \right)^{1-\alpha} \left(\frac{C_j}{N_j} \right)^{(1-\alpha)(\varepsilon_a - 1)} \tag{A.8}$$

Since $E_j = \sum_{b=\{o,a\}} P_j^b \; C_j^b$, it implies that $\sum_{b=\{o,a\}} \mathbf{s}_j^b = 1$. The average cost of real consumption at the country level P_j measures changes in the cost of living in terms of price and expenditure share and is given by :

$$P_j = \left[\sum_{b=\{o,a\}} \beta_b \; \left(P_j^b \right)^{1-\alpha} \; \left(\frac{C_j}{N_j} \right)^{(1-\alpha)(\varepsilon_b - 1)} \right]^{\frac{1}{1-\alpha}} \tag{A.9}$$

As in Comin et al. (2021), I also express the price index P_j in terms of observables and demand parameters (proof in Appendix B). The homothetic CES specification corresponds to the special case of equation (A.9) in which $\varepsilon_b = 1 \quad \forall b \in \{o, a\}$.

Appendix B. Price Index with Non-Homothetic CES Preferences

The non-homothetic CES price index depends on the consumer's utility level, which is generally unobservable. To address this challenge, I rely on expenditure shares, which are observable at the country level. The consumer's utility level is implicitly derived from the following equation:

$$\sum_{b=\{o,a\}} (\beta_b)^{\frac{1}{\alpha}} \left(\frac{C_j}{N_j} \right)^{\varepsilon_b \frac{(1-\alpha)}{\alpha}} \left(\frac{C_j^b}{N_j} \right)^{\frac{\alpha-1}{\alpha}} = 1 \tag{B.1}$$

Such that:

$$\mathbf{s}_j^b = \beta_b \left(\frac{P_j^b}{P_j} \right)^{1-\alpha} \left(\frac{C_j}{N_j} \right)^{(1-\alpha)(\varepsilon_b - 1)} , \tag{B.2}$$

and

$$P_j^{1-\alpha} = \frac{\beta_b}{\mathbf{s}_j^b} \left(P_j^b \right)^{1-\alpha} \left(\frac{C_j}{N_j} \right)^{(1-\alpha)(\varepsilon_b - 1)} \tag{B.3}$$

Taking logarithms yields:

$$\log P_j = \frac{1}{1-\alpha} \left[\log \beta_b - \log \mathbf{s}_j^b \right] + \log P_j^b + (\varepsilon_b - 1) \log \left(\frac{E_j}{P_j \ N_j} \right) \tag{B.4}$$

Applying the sum, we have:

$$\sum_b \log P_j = \frac{1}{1-\alpha} \sum_b \left[\log \beta_b - \log \mathbf{s}_j^b \right] + \sum_b \log P_j^b + \sum_b (\varepsilon_b - 1) \log \left(\frac{E_j}{P_j \ N_j} \right) \tag{B.5}$$

Averaging across sectors gives:

$$\log P_j = \frac{1}{1-\alpha} \log \tilde{\beta} - \frac{1}{1-\alpha} \log \tilde{\mathbf{s}}_j + \log \tilde{P}_j + \psi \log \left(\frac{E_j}{P_j \ N_j} \right) , \tag{B.6}$$

where a tilde above each variable denotes an average across sectors such that $\tilde{P}_j = \left(\prod_{b \in B} P_j^b \right)^{\frac{1}{\texttt{Card}(\mathbf{B})}}$, $\psi = \frac{1}{\texttt{Card}(\mathbf{B})} \sum_b (\varepsilon_b - 1)$, and $\mathbf{B}$ is the set of elements indexed by b, or the total number of sectors. Therefore:

$$(1+\psi) \log P_j = \frac{1}{1-\alpha} \log \tilde{\beta} - \frac{1}{1-\alpha} \log \tilde{\mathbf{s}}_j + \log \tilde{P}_j + \psi \log \left(\frac{E_j}{N_j} \right) \tag{B.7}$$

$$P_j = \tilde{\beta}^{\frac{1}{(1+\psi)(1-\alpha)}} \tilde{P}_j^{\frac{1}{(1+\psi)}} \tilde{\mathbf{s}}_j^{-\frac{1}{(1+\psi)(1-\alpha)}} \left(\frac{E_j}{N_j} \right)^{\frac{\psi}{1+\psi}} , \tag{B.8}$$

Appendix C. The demand for varieties

The Engel equation (A.8) determines how much each country spends on agri-food products. This expenditure is then divided across crops and, for each crop, across varieties from different origins. The agricultural aggregate is a CES aggregator of crops, with elasticity of substitution κ and weights β_{jk}, and each crop is a CES aggregator of varieties from different origins, with elasticity of substitution σ and weights β_{ijk}. The corresponding demands for crop k and for variety (i,k) are:

$$C_{jk} = \beta_{jk} P_{jk}^{-\kappa} \left(P_j^a\right)^{\kappa} C_j^a, \qquad C_{ijk} = \beta_{ijk} p_{ijk}^{-\sigma} P_{jk}^{\sigma} C_{jk}, \tag{C.1}$$

where $P_{jk} = \left[\sum_i \beta_{ijk} p_{ijk}^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$ and $P_j^a = \left[\sum_k \beta_{jk} P_{jk}^{1-\kappa}\right]^{\frac{1}{1-\kappa}}$. Expenditure on variety (i,k) in country j can be written as a product of expenditure shares:

$$p_{ijk}C_{ijk} = \frac{p_{ijk}C_{ijk}}{P_{jk}C_{jk}} \times \frac{P_{jk}C_{jk}}{P_j^a C_j^a} \times \frac{P_j^a C_j^a}{P_j C_j} \times P_j C_j. \tag{C.2}$$

Using (C.1), the share of origin i in the expenditure on crop k and the share of crop k in the expenditure on agri-food products are:

$$\frac{p_{ijk}C_{ijk}}{P_{jk}C_{jk}} = \beta_{ijk}\left(\frac{p_{ijk}}{P_{jk}}\right)^{1-\sigma}, \qquad \frac{P_{jk}C_{jk}}{P_j^a C_j^a} = \beta_{jk}\left(\frac{P_{jk}}{P_j^a}\right)^{1-\kappa}. \tag{C.3}$$

The share of agri-food products in total expenditure, $\mathbf{s}_j^a$, is given by (A.8). Substituting (C.3) and (A.8) into (C.2), we obtain the bilateral demand:

$$p_{ijk}C_{ijk} = \beta_a \left(\frac{\beta_{ijk}\, p_{ijk}^{1-\sigma}}{P_{jk}^{1-\sigma}}\right)\left(\frac{\beta_{jk}\, P_{jk}^{1-\kappa}}{(P_j^a)^{1-\kappa}}\right)\left(\frac{P_j^a}{P_j}\right)^{1-\alpha}\left(\frac{C_j}{N_j}\right)^{(1-\alpha)(\varepsilon_a-1)} P_j C_j, \tag{C.4}$$

which measures expenditure on variety (i,k) in country j. When $\varepsilon_a = 1$, the term $(C_j/N_j)^{(1-\alpha)(\varepsilon_a-1)}$ equals one, and the bilateral demand reduces to that of Costinot et al. (2016).

Appendix D. Producer Problem and Land Rents

I consider a parcel ω located in field f of country i. If this parcel is allocated to crop k, production is given by:

$$Q_i^{fk}(\omega) = A_i^{fk}(\omega) \min\left\{L_i^{fk}(\omega), \frac{N_i^{fk}(\omega)}{\nu_i^f(\omega)}\right\}, \tag{D.1}$$

where $L_i^{fk}(\omega)$ denotes land used on the parcel, $N_i^{fk}(\omega)$ denotes labor, $A_i^{fk}(\omega)$ is parcel-level

productivity, and $\nu_i^f(\omega)$ is the labor requirement. Land and labor are perfect complements.

The producer's optimization problem is to maximize land rent. For a given crop k, the producer chooses labor and land to maximize the surplus generated by the parcel:

$$\max_{L_i^{fk}(\omega),\, N_i^{fk}(\omega)} \left\{ p_i^k Q_i^{fk}(\omega) - w_i N_i^{fk}(\omega) \right\}. \tag{D.2}$$

Since the production technology is Leontief, it is never optimal to use excess labor relative to land. Therefore, whenever the parcel is operated, we have:

$$L_i^{fk}(\omega) = \frac{N_i^{fk}(\omega)}{\nu_i^f(\omega)}, \tag{D.3}$$

or equivalently,

$$N_i^{fk}(\omega) = \nu_i^f(\omega) L_i^{fk}(\omega). \tag{D.4}$$

Substituting equation (D.4) into (D.1), output becomes

$$Q_i^{fk}(\omega) = A_i^{fk}(\omega) L_i^{fk}(\omega). \tag{D.5}$$

The surplus generated by crop k on parcel ω is therefore

$$\Pi_i^{fk}(\omega) = p_i^k A_i^{fk}(\omega) L_i^{fk}(\omega) - w_i \nu_i^f(\omega) L_i^{fk}(\omega). \tag{D.6}$$

Equivalently,

$$\Pi_i^{fk}(\omega) = \left[p_i^k A_i^{fk}(\omega) - w_i \nu_i^f(\omega) \right] L_i^{fk}(\omega). \tag{D.7}$$

The term in brackets is the land rent per unit of land associated with crop k on parcel ω:

$$R_i^{fk}(\omega) = p_i^k A_i^{fk}(\omega) - w_i \nu_i^f(\omega). \tag{D.8}$$

Thus, the return to land is not an exogenous cost. It is the residual value of production after paying labor costs.

In the non-agricultural sector, I assume that labor is the only factor of production:

$$Q_i^o = A_i^o N_i^o, \tag{D.9}$$

where A_i^o denotes labor productivity in the non-agricultural sector and N_i^o is labor employed in that sector. The representative firm in the outside sector solves:

$$\max_{N_i^o} \Pi_i^o = P_i^o Q_i^o - w_i N_i^o. \tag{D.10}$$

Substituting equation (D.9) into (D.10) gives

$$\Pi_i^o = \left(P_i^o A_i^o - w_i\right) N_i^o. \tag{D.11}$$

Under perfect competition and constant returns to scale, equilibrium profits are zero whenever the outside good is produced. Hence,

$$P_i^o A_i^o - w_i = 0, \tag{D.12}$$

which implies

$$w_i = P_i^o A_i^o. \tag{D.13}$$

Since the outside good is freely traded and used as the numeraire, its price is normalized to one:

$$P_i^o = 1. \tag{D.14}$$

Therefore,

$$w_i = A_i^o. \tag{D.15}$$

This condition means that the wage is pinned down by labor productivity in the freely traded non-agricultural sector.

Plugging (D.15) into (D.8), the land rent can be written as

$$R_i^{fk}(\omega) = p_i^k A_i^{fk}(\omega) - A_i^o \nu_i^f(\omega). \tag{D.16}$$

Appendix E. Crop Choice and Land Allocation

The producer allocates the parcel to the crop that generates the highest non-negative land rent. Since $\nu_i^f(\omega)$ does not vary across crops within a given parcel, the labor-cost term is common across crops. Therefore, conditional on cultivation, the crop choice depends on the value of marginal product, $p_i^k A_i^{fk}(\omega)$.

A parcel is cultivated with crop k only if the value of producing crop k on that parcel exceeds both the value of producing any other crop on the same parcel and the labor cost required to operate the parcel. Formally, this requires:

$$p_i^k A_i^{fk}(\omega) > p_i^\ell A_i^{f\ell}(\omega) \qquad \forall \ell \neq k, \tag{E.1}$$

and

$$p_i^k A_i^{fk}(\omega) > A_i^o \nu_i^f(\omega). \tag{E.2}$$

Equivalently, crop k is chosen on parcel ω if and only if

$$p_i^k A_i^{fk}(\omega) = \max\left\{A_i^o \nu_i^f(\omega), p_i^1 A_i^{f1}(\omega), \ldots, p_i^K A_i^{fK}(\omega)\right\}. \tag{E.3}$$

The share of field f allocated to crop k is therefore

$$\pi_i^{fk} = \Pr\left[p_i^k A_i^{fk}(\omega) = \max\left\{A_i^o \nu_i^f(\omega), p_i^1 A_i^{f1}(\omega), \ldots, p_i^K A_i^{fK}(\omega)\right\}\right]. \tag{E.4}$$

Following Costinot et al. (2016), parcel-level productivity $A_i^{fk}(\omega)$ and labor requirements $\nu_i^f(\omega)$ are unobserved and independently drawn from Fréchet distributions with a common shape parameter $\theta > 1$. Their scale parameters are chosen so that the means are $A_i^{fk} = E\left[A_i^{fk}(\omega)\right]$, provided by the GAEZ data, and $\nu_i = E\left[\nu_i^f(\omega)\right]$, the average labor requirement in agriculture. The labor-cost term is summarized by the country-specific parameter

$$a_i \equiv A_i^o \nu_i. \tag{E.5}$$

Since the mean of a Fréchet random variable is proportional to its scale, with a factor $\Gamma(1-1/\theta)$ common to all draws, the probability in (E.4) becomes

$$\pi_i^{fk} = \frac{\left(p_i^k A_i^{fk}\right)^\theta}{a_i^\theta + \sum_{\ell \in \mathcal{K}} \left(p_i^\ell A_i^{f\ell}\right)^\theta}. \tag{E.6}$$

The supply side is disciplined by two parameters: the within-field productivity dispersion parameter θ and the country-specific labor-cost shifter a_i. The former governs the elasticity of land allocation with respect to crop prices and average productivity, while the latter controls the extent to which land is profitably cultivated. These parameters are calibrated in the empirical implementation.

Appendix F. Supply of Crops

The Leontief specification indicates that, conditional on the parcel being operated, output is equal to land productivity multiplied by the cultivated area. The aggregate output of crop k in country i is obtained by summing production over all fields.

$$Q_i^k = \sum_f \int_0^1 Q_i^{fk}(\omega) d\omega \tag{F.1}$$

For a given field f, let s_i^f denote the field's area and π_i^{fk} the share of that field allocated to crop k. Production in field f for crop k is therefore given by the amount of land allocated to crop k, $s_i^f \pi_i^{fk}$, multiplied by the average productivity of parcels selected into crop k:

$$Q_i^{fk} = s_i^f \pi_i^{fk} E\left[A_i^{fk}(\omega) \mid k \text{ is chosen}\right]. \tag{F.2}$$

Under the Fréchet assumption, the expected productivity of parcels allocated to crop k is

$$E\left[A_i^{fk}(\omega) \mid k \text{ is chosen}\right] = A_i^{fk} \left(\pi_i^{fk}\right)^{-1/\theta}. \tag{F.3}$$

Substituting equation (F.3) into (F.2) gives

$$Q_i^{fk} = s_i^f \pi_i^{fk} A_i^{fk} \left(\pi_i^{fk}\right)^{-1/\theta}. \tag{F.4}$$

Therefore,

$$Q_i^{fk} = s_i^f A_i^{fk} \left(\pi_i^{fk}\right)^{\frac{\theta-1}{\theta}}. \tag{F.5}$$

Using equation (F.5), (F.1) becomes

$$Q_i^k = \sum_{f \in \mathcal{F}_i} s_i^f A_i^{fk} \left(\pi_i^{fk}\right)^{\frac{\theta-1}{\theta}}. \tag{F.6}$$

Finally, substituting the land-allocation share equation (E.6) into equation (F.6) yields:

$$Q_i^k = \sum_{f \in \mathcal{F}_i} s_i^f A_i^{fk} \left[\frac{\left(p_i^k A_i^{fk}\right)^\theta}{a_i^\theta + \sum_{\ell \in \mathcal{K}} \left(p_i^\ell A_i^{f\ell}\right)^\theta}\right]^{\frac{\theta-1}{\theta}}. \tag{F.7}$$

Appendix G. Sample Coverage

Table G.1: Crops and Share of World Crop Output Value

Crop	Share of world crop output (%)
Rice	16.83
Maize	11.03
Wheat	10.38
Cotton	6.47
Tomato	5.70
White Potato	5.47
Soybean	5.44
Sugarcane	4.52
Banana	3.08
Oil Palm	2.87
Total	71.79

Source: FAOSTAT. Shares computed over the 50-country sample.

Table G.2: Countries by Income Group and Share of World Crop Output Value

Low income		Lower-middle income		Upper-middle income		High income	
Country	%	Country	%	Country	%	Country	%
Congo, Dem. Rep.	0.41	Burma	1.63	China	22.13	Poland	0.40
Malawi	0.41	Bangladesh	0.78	Thailand	1.28	Korea, Rep.	1.00
Ethiopia	0.53	Viet Nam	1.92	Ecuador	0.40	Greece	0.79
Tanzania	0.58	Ghana	0.48	Algeria	0.41	Spain	1.14
Uganda	0.70	Uzbekistan	0.75	Colombia	0.63	United Kingdom	0.50
		Pakistan	1.23	Iran	0.82	Canada	1.14
		India	9.21	Kazakhstan	0.36	Japan	2.92
		Cameroon	0.40	South Africa	0.38	Germany	1.07
		Côte d'Ivoire	0.40	Malaysia	0.59	Australia	1.03
		Nigeria	3.04	Russia	1.99	Italy	0.96
		Sudan	0.41	Brazil	4.87	United States	9.05
		Philippines	0.82	Argentina	1.44	Netherlands	0.40
		Indonesia	4.05	Romania	0.44	France	1.19
		Ukraine	0.81	Mexico	0.90		
		Egypt	0.95	Venezuela	0.71		
		Morocco	0.37	Turkey	2.35		

Notes: Income group classification follows the World Bank classification. *Source:* FAOSTAT and World Bank data.

Table G.3: Climate-Induced Productivity Shocks and Agricultural Expenditure Shares by Country

Country	Yield change (%)	ΔA_i (% of GDP)	s_j^a (%)
Low income			
Congo, Dem. Rep.	-45.9	-7.04	14.5†
Ethiopia	-31.5	-3.75	41.0
Malawi	-51.3	-38.41	31.1
Tanzania	-30.2	-5.91	31.7
Uganda	-11.3	-3.82	15.1
Lower-middle income			
Bangladesh	-27.8	-3.44	28.4
Burma	-12.0	-5.96	37.4†
Côte d'Ivoire	-34.5	-2.52	36.6
Cameroon	-32.7	-3.07	27.9
Egypt	-86.6	-1.96	13.3
Ghana	-59.5	-6.37	35.7
India	-24.9	-1.84	20.8
Indonesia	-15.5	-1.26	12.8
Morocco	-52.5	-1.57	25.0
Nigeria	-32.6	-2.59	60.0
Pakistan	-20.4	-1.74	17.5
Philippines	-23.3	-1.32	13.4
Sudan	-86.2	-3.35	25.0†
Ukraine	-9.9	-0.54	10.7
Uzbekistan	-10.8	-2.22	21.1†
Viet Nam	-18.5	-4.16	15.7
Upper-middle income			
Algeria	-35.0	-0.88	20.9†
Argentina	4.8	0.20	5.0
Brazil	-60.8	-1.94	5.9
China	-10.1	-0.46	8.8
Colombia	-62.2	-1.47	7.9
Ecuador	-55.9	-4.61	16.1
Iran	-39.6	-0.94	12.6
Kazakhstan	-14.5	-0.49	16.7
Malaysia	-7.3	-0.35	3.8
Mexico	-36.5	-0.39	4.1
Romania	-21.7	-0.60	22.7
Russia	-1.6	-0.03	11.5
South Africa	-43.6	-0.54	6.8
Thailand	-43.2	-2.61	8.4
Turkey	-10.8	-0.30	9.2
Venezuela	-89.5	-2.24	9.0
High income			
Australia	-28.7	-0.23	1.3
Canada	43.4	0.26	2.2
France	-23.2	-0.08	3.1
Germany	6.1	0.01	2.0
Greece	-11.3	-0.13	3.8
Italy	-17.5	-0.07	3.8
Japan	7.0	0.04	2.4
Korea, Rep.	12.1	0.17	4.8
Netherlands	34.3	0.08	2.6
Poland	9.3	0.06	6.5
Spain	-21.3	-0.09	3.1
United Kingdom	-0.8	-0.00	2.4
United States	-16.7	-0.14	2.0

Notes: Yield change is the average change in field-level productivity between the baseline and future climates, weighted by observed crop output. ΔA_i is the weighted average productivity change, expressed in percent of GDP. s_j^a is the share of household consumption spent on agricultural goods. † Not available as a separate region in GTAP 9. The share is that of the composite region that contains the country: Rest of Southeast Asia (Burma), Rest of Eastern Africa (Sudan), Rest of Former Soviet Union (Uzbekistan), Rest of North Africa (Algeria), and South Central Africa (Congo, Dem. Rep.). *Source:* GAEZ, FAOSTAT, and GTAP.

Appendix H. Welfare Measurement

Let $e(p, u)$ denote the expenditure function derived from the non-homothetic CES preferences, p^0 and p^1 the price vectors before and after the climate shock, u^0 and u^1 the corresponding utility levels, and I^0 and I^1 the level of income before and after the shock.

Equivalent variation

Following the original definition of Hicks (1946), as clarified by Diewert (1992), the equivalent variation evaluates the change in utility at pre-shock prices:

$$EV = e(p^0, u^1) - e(p^0, u^0). \tag{H.1}$$

Diewert (1992) notes that Hicks (1946) later used an alternative definition, which evaluates the change in prices at the post-shock utility level:

$$H = e(p^1, u^1) - e(p^0, u^1). \tag{H.2}$$

Relation between the two measures

Under cost minimization, the minimum expenditure required to attain the utility level actually achieved equals income in each period:

$$e(p^0, u^0) = I^0, \qquad e(p^1, u^1) = I^1. \tag{H.3}$$

Substituting (H.3) into (H.1), then adding and subtracting $e(p^1, u^1) = I^1$, gives

$$EV = e(p^0, u^1) - e(p^1, u^1) + \left(I^1 - I^0\right) = -H + \left(I^1 - I^0\right). \tag{H.4}$$

When income is unchanged ($I^1 = I^0$), $EV = -H$: the two measures have the same magnitude and opposite signs. H is the compensation required to reach u^1 at post-shock rather than pre-shock prices, which is positive when prices rise, whereas EV measures the welfare change itself, which is negative when prices rise. EV also captures the change in income, which H does not.

Welfare change in percent of income

In the quantitative analysis, income I^1 equals GDP plus the change in net farm income and in trade-tax revenue. I report the equivalent variation relative to baseline income:

$$\Delta W = \frac{EV}{e(p^0, u^0)} = \frac{e(p^0, u^1)}{e(p^0, u^0)} - 1. \tag{H.5}$$

Decomposition

Using equation (H.5), I multiply and divide by $e(p^1, u^1)$:

$$1 + \Delta W = \frac{e(p^0, u^1)}{e(p^0, u^0)} = \frac{e(p^1, u^1)}{e(p^0, u^0)} \times \frac{e(p^0, u^1)}{e(p^1, u^1)} = \frac{I^1}{I^0} \times \frac{e(p^0, u^1)}{e(p^1, u^1)}, \tag{H.6}$$

where the last equality uses equation (H.3). The first term is the change in income. The second term is the ratio of the cost of living before and after the shock, evaluated at the post-shock utility level u^1. Multiplying and dividing this second term by the same ratio evaluated at the baseline utility level, $e(p^0, u^0)/e(p^1, u^0)$, yields

$$1 + \Delta W = \underbrace{\frac{I^1}{I^0}}_{\text{income effect}} \times \underbrace{\frac{e(p^0, u^0)}{e(p^1, u^0)}}_{\text{price effect}} \times \underbrace{\frac{e(p^0, u^1)/e(p^1, u^1)}{e(p^0, u^0)/e(p^1, u^0)}}_{\text{structural change effect}}. \tag{H.7}$$

The income effect captures the change in income. The price effect measures the change in the cost of living induced by the shock at the baseline utility level u^0. It also measures how much more it costs to maintain the previous standard of living at the new food prices, and serves as a measure of the cost of living. The structural change effect captures how this cost-of-living ratio changes when utility moves from u^0 to u^1. Under homothetic preferences ($\varepsilon_a = \varepsilon_o = 1$), the ratio $e(p^0, u)/e(p^1, u)$ does not depend on u, so that the structural change effect equals one. With non-homothetic preferences, it captures how the change in utility shifts the share of agricultural goods in consumption. In the quantitative analysis, each component is reported in percent of income.

Table H.1: Decomposition of the Welfare Effects by Country (% of Income)

Country	Agri. price	Price effect	Struct. change	Income effect	Country	Agri. price	Price effect	Struct. change	Income effect
Low income					Upper-middle income				
Congo, Dem. Rep.	130.0	−13.64	−0.74	2.90	Algeria	28.2	−5.31	−0.13	0.06
Ethiopia	10.1	−3.93	−0.06	0.73	Argentina	−2.4	0.12	0.00	0.62
Malawi	122.9	−26.17	−1.95	12.03	Brazil	102.2	−4.78	−0.10	0.25
Tanzania	40.0	−10.75	−0.48	1.59	China	−2.0	0.17	0.00	0.06
Uganda	16.5	−2.37	−0.01	1.45	Colombia	87.4	−5.54	−0.16	−0.28
					Ecuador	14.8	−2.26	−0.02	0.44
Lower-middle income					Iran	22.8	−2.67	−0.03	0.08
Bangladesh	87.5	−18.00	−1.65	0.80	Kazakhstan	34.7	−5.17	−0.12	0.22
Burma	31.8	−10.33	−0.37	2.67	Malaysia	42.6	−1.45	−0.01	−0.10
Côte d'Ivoire	48.0	−14.04	−1.00	−0.13	Mexico	33.3	−1.24	−0.01	0.00
Cameroon	51.7	−11.76	−0.65	0.62	Romania	24.0	−4.98	−0.11	0.07
Egypt	6.1	−0.79	0.00	−0.19	Russia	−51.8	7.41	−0.23	−0.11
Ghana	201.3	−35.81	−8.46	1.69	South Africa	60.1	−3.51	−0.06	0.14
India	32.5	−6.00	−0.16	0.53	Thailand	194.9	−10.96	−0.63	−0.35
Indonesia	51.8	−5.68	−0.14	0.40	Turkey	−0.4	0.04	0.00	0.02
Morocco	43.7	−9.19	−0.44	−0.52	Venezuela	279.3	−14.95	−1.20	−0.11
Nigeria	112.9	−36.88	−8.14	0.81					
Pakistan	28.7	−4.58	−0.07	1.15	High income				
Philippines	29.6	−3.61	−0.07	−0.47	Australia	39.3	−0.45	0.00	0.14
Sudan	341.7	−36.28	−9.79	0.09	Canada	−0.5	0.01	0.00	0.76
Ukraine	6.7	−0.70	0.00	0.20	France	4.8	−0.15	0.00	−0.02
Uzbekistan	−8.2	1.79	−0.01	−0.28	Germany	4.3	−0.08	0.00	0.03
Viet Nam	34.3	−4.82	−0.11	0.15	Greece	−4.4	0.17	0.00	−0.04
					Italy	1.0	−0.04	0.00	−0.03
					Japan	−1.8	0.04	0.00	0.02
					Korea, Rep.	23.7	−1.06	−0.01	0.02
					Netherlands	−12.0	0.32	0.00	0.08
					Poland	−13.3	0.91	0.00	0.05
					Spain	−13.0	0.42	0.00	−0.08
					United Kingdom	−25.2	0.65	0.00	0.11
					United States	15.1	−0.29	0.00	0.02

Notes: Hadley CM3 A1FI scenario, non-homothetic preferences with $\varepsilon_a = 0.083$. Agri. price: percent change, between the baseline and counterfactual equilibria, in the consumer price index of agricultural goods, expressed relative to the outside good. Increases above 100% occur in countries where crop yields fall by more than 30% and where most crop consumption is domestically produced at baseline. The price, structural change (Struct.) and income effects are in percent of the baseline income. Income groups follow the World Bank classification. *Source:* Author's calculations based on GAEZ, FAOSTAT, GTAP and WDI data.

Table H.2: Welfare Effects of Climate Change with Adaptation (% of Income)

	Low	Lower-middle	Upper-middle	High	World
A. Welfare change					
No adaptation (benchmark)	−6.23	−8.53	−1.09	−0.04	−0.83
Adaptation in lower-income countries	−1.76	−4.08	−1.04	−0.01	−0.51
Adaptation in all countries	−1.72	−3.82	0.38	0.17	−0.04
B. Agricultural price change					
No adaptation (benchmark)	40.9	45.6	28.1	5.5	13.4
Adaptation in lower-income countries	14.0	19.6	26.9	4.0	10.4
Adaptation in all countries	13.5	17.3	1.8	−5.0	−2.0
C. Income effect					
No adaptation (benchmark)	1.99	0.42	0.06	0.05	0.08
Adaptation in lower-income countries	0.84	0.23	0.04	0.04	0.05
Adaptation in all countries	0.73	0.12	−0.04	0.01	0.01

Notes: Hadley CM3 A1FI scenario, non-homothetic preferences with $\varepsilon_a = 0.083$; GDP-weighted averages within each income group. Adaptation reduces climate-induced declines in potential yields by 50% for each field and crop; yield gains remain unchanged. Adaptation in lower-income countries applies to low-income and lower-middle-income countries only. Panel A: equivalent variation in percent of baseline income. Panel B: percent change in the consumer price index of agricultural goods. Panel C: income effect of the welfare decomposition, in percent of baseline income. Income groups follow the World Bank classification. *Source:* Author's calculations based on GAEZ, FAOSTAT, GTAP and WDI data.

Appendix I. Sensitivity

Table I.1: Sensitivity of Welfare Effects to the Elasticity of Substitution (% of Income)

		World (GDP-weighted)		Structural	Income group			
α	ε_a	Non-homothetic	Homothetic	effect	Low	Lower-middle	Upper-middle	High
0.20	0.351	−1.05	−0.92	−0.11	−6.68	−9.56	−1.63	−0.08
0.26	0.318	−1.00	−0.88	−0.10	−6.57	−9.29	−1.50	−0.07
0.30	0.292	−0.96	−0.85	−0.10	−6.50	−9.13	−1.42	−0.06
0.40	0.208	−0.89	−0.79	−0.09	−6.35	−8.77	−1.24	−0.05
0.50*	0.083	−0.83	−0.74	−0.09	−6.23	−8.53	−1.09	−0.04

Notes: Equivalent variation in percent of baseline income under the Hadley CM3 A1FI scenario. For each value of α, ε_a is estimated by nonlinear least squares on the relative expenditure share equation, and $\beta_{a,j}$ is recalibrated by country. The homothetic case sets $\varepsilon_a = \varepsilon_o = 1$ for the same value of α. The structural effect is the GDP-weighted structural change component of the welfare decomposition. Income groups follow the World Bank classification; income-group columns are GDP-weighted averages under non-homothetic CES preferences. * Benchmark specification. For $\alpha = 0.6$ the estimated ε_a is negative and the counterfactual equilibrium is not solved. *Source:* Author's calculations using GAEZ, FAOSTAT, GTAP and WDI data.

Appendix J. Trade Policy Scenarios

Table J.1 lists the three largest exporters of each crop, which impose export taxes in the export restriction scenarios. Table J.2 separates the effect of each policy under climate shocks from its effect without climate shocks.

Table J.1: Three Largest Exporters by Crop

Crop	First exporter	Second exporter	Third exporter
Banana	Ecuador (28.4%)	Colombia (15.9%)	Philippines (12.8%)
Cotton	United States (40.4%)	India (10.7%)	Brazil (8.1%)
Maize	United States (45.3%)	France (8.8%)	Argentina (8.4%)
Oil palm	Indonesia (41.9%)	Malaysia (41.6%)	Netherlands (4.6%)
Rice	United States (57.3%)	China (8.0%)	India (7.1%)
Soybean	United States (44.8%)	Brazil (38.2%)	Argentina (6.6%)
Sugarcane	China (34.3%)	Egypt (6.0%)	Morocco (4.5%)
Tomato	Netherlands (19.7%)	Mexico (18.0%)	Spain (17.9%)
Wheat	Canada (16.1%)	United States (14.3%)	France (12.4%)
White potato	Netherlands (22.3%)	France (15.3%)	Germany (7.9%)

Notes: Exporters ranked by the value of observed bilateral exports, excluding domestic sales; shares of total exports of each crop within the sample in parentheses. Trade is recorded for raw products (e.g., paddy rice, raw sugarcane), which explains why some major exporters of processed products (e.g., Thailand for milled rice, Russia for wheat) do not appear. *Source:* FAOSTAT trade data as compiled by Costinot et al. (2016).

Table J.2: Trade Policy Effects on welfare With and Without Climate Change

	World			Low income			Lower-middle income		
Policy	Total	Policy only	Climate-specific	Total	Policy only	Climate-specific	Total	Policy only	Climate-specific
30% export tax removed[a]	0.09	0.08	0.01	−0.13	0.02	−0.15	0.37	0.12	0.25
Top-3 exporters, 25%	−0.04	−0.04	0.00	−0.04	−0.03	−0.01	−0.18	−0.06	−0.12
Top-3 exporters, 50%	−0.08	−0.07	−0.01	−0.06	−0.05	−0.01	−0.33	−0.14	−0.19
Insulation, all crops, 50%	−0.05	0.00	−0.05	−0.11	0.00	−0.11	−0.19	0.00	−0.19
Insulation, all crops, 100%	−0.10	0.00	−0.10	−0.14	0.00	−0.14	−0.40	0.00	−0.40
Insulation, staples, 50%	−0.01	0.00	−0.01	−0.04	0.00	−0.04	0.04	0.00	0.04
Insulation, staples, 100%	−0.02	0.00	−0.02	−0.07	0.00	−0.07	0.00	0.00	0.00

Notes: Changes in the equivalent variation, in percentage points of baseline income, GDP-weighted averages. Total: welfare under climate shocks (Hadley CM3 A1FI) with the policy, minus welfare under climate shocks without the policy. Policy only: welfare effect of the same policy without climate shocks. Climate-specific: difference between the two, which measures how the effect of the policy depends on climate change. Insulation policies are triggered only by a rise in domestic prices, so that their policy-only effect is zero. [a] Relative to a baseline with a 30% export tax on all crops; the comparison is with climate shocks under the maintained tax. Income groups follow the World Bank classification. *Source:* Author's calculations based on GAEZ, FAOSTAT, GTAP and WDI data.